

Solutions to Dummit & Foote's Abstract Algebra 3rd Edition

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Preface

This collection contains solutions to the exercises in David S. Dummit and Richard M. Foote’s *Abstract Algebra*, Third Edition—a comprehensive graduate-level text that has become a standard reference in the field. I have worked through these problems over time to deepen my understanding of the material and to create a resource for others studying abstract algebra. While these solutions are written with care and rigor, they are not guaranteed to be error-free; if you discover mistakes or have suggestions for improvement, I welcome corrections.

Solutions are organized by chapter and section, mirroring the structure of the textbook. Within each section, exercises are presented in the order they appear in the book. Each solution aims to be self-contained but frequently references prior results; when a proof depends heavily on another exercise or theorem, I note this explicitly.

Exercise 0.0.1

Most problems in the book are presented in the standard exercise environment. These exercises are typically routine and don’t require advanced mathematical insight beyond the techniques and concepts introduced in the text.

Exercise 0.0.2

Problems in the “special exercise” environment are included because they are particularly important, instructive, or deep. Some are challenging in their proof techniques or conceptual demands; others are simple but illustrate key ideas fundamental to the overall text. These exercises deepen understanding and often provide insight into material developed later in the book.

This is an ongoing project. As I work through the text, I refine earlier solutions and add new ones. While I strive for mathematical correctness, errors may remain—particularly in computational problems or long proofs. I encourage you to read critically, verify claims, and develop your own solutions. The goal is not to provide answers to be memorized, but to illustrate good problem-solving techniques and deepen your understanding of abstract algebra.

I am completing this project independently at my own pace. That said, I genuinely welcome corrections, suggestions, and alternative solutions from the community. If you find errors, have ideas for improvement, or wish to contribute, please open an issue or submit a pull request on GitHub. You may also reach out directly via Reddit or Discord with feedback or discussion. I appreciate the interest in helping this project grow and remain open to community involvement.

My deepest thanks to David S. Dummit and Richard M. Foote for writing *Abstract Algebra*. Their exposition is remarkably clear, their choice of exercises is thoughtful and comprehensive, and their book has been instrumental in my mathematical development. Many of these solutions exist because of the beauty and depth of their presentation.

Finally, I’d like to take a moment and thank the broader mathematics community for maintaining a culture of shared knowledge. The practice of working through textbooks and sharing solutions has a long and valuable tradition, and I am grateful to everyone who contributes to this collective effort. I hope that this project, in its small way, adds to that tradition and proves useful to others in their study of abstract algebra. Please enjoy!

Contributors

As of now, I am independently forming solutions to each exercise myself. However, I welcome any corrections, suggestions, or slicker solutions from others. A (growing) list of contributors will be maintained here. If you'd like to contact me, please reach out on Reddit at [my profile](#) or via Discord, where my username is blanketism. Thank you to everyone who decided to take time out of their busy lives to contribute to this project. I appreciate your help in making it better for everyone.

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0 Preliminaries

0.1 Basics

For Exercise 0.1.1 through Exercise 0.1.4, let

$$M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

and $\mathcal{B} = \{X \in \mathcal{A} \mid MX = XM\}$, where $\mathcal{A}$ denotes the set of 2×2 matrices with real entries.

Exercise 0.1.1

Determine which of the following elements of $\mathcal{A}$ lie in $\mathcal{B}$:

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Solution. Let M_i denote the i -th matrix in the list. M_1, M_3, M_5, M_6 is M , the zero matrix, I_2 , and the exchange matrix respectively. Among these, M_1, M_3, M_5 belong to $\mathcal{B}$, while $M_6 \notin \mathcal{B}$. Direct calculation shows M_2 and M_4 are not in $\mathcal{B}$. ■

Exercise 0.1.2

Prove that if $P, Q \in \mathcal{B}$, then $P + Q \in \mathcal{B}$ where $+$ denotes the usual sum of two matrices.

Solution. If $P, Q \in \mathcal{B}$, then $M(P + Q) = MP + MQ = PM + QM = (P + Q)M$, so $P + Q \in \mathcal{B}$. ■

Exercise 0.1.3

Prove that if $P, Q \in \mathcal{B}$, then $P \cdot Q \in \mathcal{B}$ where $\cdot$ denotes the usual product of two matrices.

Solution. If $P, Q \in \mathcal{B}$, then $M(PQ) = (MP)Q = (PM)Q = P(MQ) = P(QM) = (PQ)M$, so $PQ \in \mathcal{B}$. ■

Exercise 0.1.4

Find conditions on p, q, r, s which determine precisely when

$$\begin{pmatrix} p & q \\ r & s \end{pmatrix} \in \mathcal{B}.$$

Solution. Let X be the matrix above. Computing $MX = XM$ gives the following system of equations:

$$\begin{cases} p + r = p \\ q + s = p + q \\ r = r \\ s = r + s \end{cases}$$

Simplifying results in $r = 0$ and $p = s$. Then

$$\mathcal{B} = \left\{ \begin{pmatrix} p & q \\ 0 & p \end{pmatrix} \mid p, q \in \mathbb{R} \right\}.$$

Exercise 0.1.5

Determine whether the following functions f are well-defined:

- (a) $f : \mathbb{Q} \rightarrow \mathbb{Z}$ defined by $f(a/b) = a$.
- (b) $f : \mathbb{Q} \rightarrow \mathbb{Q}$ defined by $f(a/b) = a^2/b^2$.

Solution.

- (a) No: $1/2 = 2/4$, but $f(1/2) = 1 \neq 2 = f(2/4)$.
- (b) Yes: $a/b = c/d$ implies $a^2/b^2 = c^2/d^2$, so $f(a/b) = f(c/d)$. ■

Exercise 0.1.6

Determine whether the function $f : \mathbb{R}^+ \rightarrow \mathbb{Z}$ defined by mapping a real number r to the first digit to the right of the decimal point in a decimal expansion of r is well-defined.

Solution. No. Note that $1 = 1.000\dots = 0.999\dots$, but $f(1.000\dots) = 1$ and $f(0.999\dots) = 9$. ■

Exercise 0.1.7

Let $f : A \rightarrow B$ be a surjective map of sets. Prove that the relation

$$a \sim b \iff f(a) = f(b)$$

is an equivalence relation whose equivalence classes are the fibers of f .

Solution. That $\sim$ is an equivalence relation follows from $=$ being an equivalence relation.

The equivalence class of $a \in A$ is $\{x \in A \mid x \sim a\}$. By definition, this is $\{x \in A \mid f(x) = f(a)\}$, which is precisely the fiber of f over $f(a)$. ■

0.2 Properties of the Integers

Exercise 0.2.1

For each of the following pairs of integers a and b , determine their greatest common divisor, their least common multiple, and write their greatest common divisor in the form $ax + by$ for some integers x and y .

- (a) $a = 20, b = 13$
- (b) $a = 69, b = 372$
- (c) $a = 792, b = 275$
- (d) $a = 11391, b = 5673$
- (e) $a = 1761, b = 1567$
- (f) $a = 507885, b = 60808$

Solution.

	(a, b)	$\text{lcm}(a, b)$	$ax + by$
(a)	1	260	$2(20) - 3(13)$
(b)	3	8556	$27(69) - 5(372)$
(c)	11	19800	$8(792) - 23(275)$
(d)	3	21540381	$253(5673) - 126(11391)$
(e)	1	2759487	$-105(1761) + 118(1567)$
(f)	691	44693880	$142(60808) - 17(507885)$

Here, the lcm was computed using $ab = (a, b) \text{lcm}(a, b)$. ■

Exercise 0.2.2

Prove that if the integer k divides the integers a and b , then k divides $as + bt$ for every pair of integers s and t .

Solution. Since $k \mid a$ and $k \mid b$, there exist $x, y \in \mathbb{Z}$ such that $a = kx$ and $b = ky$. Then for any $s, t \in \mathbb{Z}$, we have

$$as + bt = (kx)s + (ky)t = k(xs + yt),$$

which shows that $k \mid (as + bt)$. ■

Exercise 0.2.3

Prove that if n is composite, then there are integers a and b such that n divides ab , but n does not divide either a or b .

Solution. Since n is composite, put $n = ab$ with $1 < a, b < n$. Then $n \mid ab$, but $n \nmid a$ and $n \nmid b$. ■

Exercise 0.2.4

Let a, b and N be fixed integers with a and b nonzero, and let $d = (a, b)$ be the greatest common divisor of a and b . Suppose x_0 and y_0 are particular solutions to $ax + by = N$. Prove, for any integer t , that the integers

$$x = x_0 + \frac{b}{d}t \quad \text{and} \quad y = y_0 - \frac{a}{d}t$$

are also solutions to $ax + by = N$.

REMARK. This is, in fact, the general solution.

Solution.

$$\begin{aligned} ax + by &= a \left(x_0 + \frac{b}{d}t \right) + b \left(y_0 - \frac{a}{d}t \right) \\ &= ax_0 + \frac{ab}{d}t + by_0 - \frac{ab}{d}t \\ &= ax_0 + by_0 = N \end{aligned}$$

■

Exercise 0.2.5

Determine the value $\varphi(n)$ for each integer $n \leq 30$, where φ denotes the Euler φ -function.

Solution.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$\varphi(n)$	1	1	2	2	4	2	6	4	6	4	10	4	12	6	8

n	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
$\varphi(n)$	8	16	6	18	8	12	10	22	8	20	12	18	12	28	8

■

Exercise 0.2.6

Prove the Well Ordering Property of $\mathbb{Z}$ by induction, and prove the minimal element is unique.

Solution. Suppose S has no minimal element, and let T be $\mathbb{Z}^+ \setminus S$, the set of nonnegative integers not in S . Since 0 is minimal in $\mathbb{Z}^+$, then $0 \notin S$, hence $0 \in T$. Fix $k \in \mathbb{Z}^+$, and suppose every j such that $0 \leq j \leq k$ is in T . Observe $k+1 \notin S$, for otherwise it would be minimal in S , since every integer less than or equal to k is in T . Then $k+1 \in T$. By induction, every nonnegative integer is in T , contradicting that S is nonempty. Therefore, S must have a minimal element.

If $s_1, s_2 \in S$ are both minimal, then $s_1 \leq s_2$ and $s_2 \leq s_1$ imply $s_1 = s_2$.

■

Exercise 0.2.7

If p is a prime, prove that there do not exist nonzero integers a and b such that $a^2 = pb^2$.

Solution. Suppose there existed a, b such that $a^2 = pb^2$. If $(a, b) \neq 1$, divide both a and b by (a, b) to get relatively prime integers. Then $p \mid a^2$, which implies $p \mid a$. Put $a = pk$ for $k \in \mathbb{Z}$. Then

$$a^2 = (pk)^2 = p^2k^2 = pb^2,$$

hence $b^2 = pk^2$. But then $p \mid b$, contradicting that $(a, b) = 1$.

■

Exercise 0.2.8

Let p be a prime, $n \in \mathbb{Z}^+$. Find a formula for the largest power of p which divides $n! = n(n-1)(n-2) \dots 2 \cdot 1$.

HINT. It involves the greatest integer function.

Solution. Note that there exists $k_1 \in \mathbb{Z}^+$ such that $n \geq k_1 p$. Then the integers $p, 2p, \dots, k_1 p$ contribute k_1 factors of p in $n!$. Next, there exists $k_2 \in \mathbb{Z}^+$ such that $n \geq k_2 p^2$. Then the integers $p^2, 2p^2, \dots, k_2 p^2$ contribute k_2 additional factors of p in $n!$. We continue this process until some $k_m \in \mathbb{Z}^+$ such that $n \geq k_m p^m$ but $n < k_{m+1} p^{m+1}$. The largest power of p which divides $n!$ is then given by

$$\sum_{i=1}^m k_i = \sum_{i=1}^m \left\lfloor \frac{n}{p^i} \right\rfloor. \quad \blacksquare$$

Exercise 0.2.9

Write a computer program to determine the greatest common divisor (a, b) of two integers a and b and to express (a, b) in the form $ax + by$ for some integers x and y .

Solution. Please see `euclidean_algorithm.py` for the solution. ■

Exercise 0.2.10

Prove for any given positive integer N there exist only finitely many integers n with $\varphi(n) = N$ where φ denotes Euler's φ -function. Conclude in particular that φ tends to infinity as n tends to infinity.

Solution. Fix $N \in \mathbb{Z}^+$, and consider $n \in \mathbb{Z}^+$ such that $\varphi(n) = N$. Let $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$ be the prime factorization of n . Then

$$\varphi(n) = \varphi(p_1^{\alpha_1}) \varphi(p_2^{\alpha_2}) \dots \varphi(p_k^{\alpha_k}) = \prod_{i=1}^k p_i^{\alpha_i-1} (p_i - 1).$$

Since $\varphi(n) = N$, there are only finitely many possibilities for the primes p_i and their exponents α_i that satisfy this equation. Hence, there are only finitely many integers n such that $\varphi(n) = N$.

If φ was finite as n tends to infinity, then there would exist some $M \in \mathbb{Z}^+$ such that $\varphi(n) \leq M$ for infinitely many n . However, this contradicts the fact that for each fixed $N \in \mathbb{Z}^+$, there are only finitely many n such that $\varphi(n) = N$. Hence, φ must tend to infinity as n tends to infinity. ■

Exercise 0.2.11

Prove that if d divides n then $\varphi(d)$ divides $\varphi(n)$ where φ denotes Euler's φ -function.

Solution. Let $n \in \mathbb{Z}^+$ with $d \mid n$. Put their prime factorizations as

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}, \quad d = p_1^{\beta_1} p_2^{\beta_2} \dots p_j^{\beta_j},$$

where $j \leq k$ and $1 \leq \beta_i \leq \alpha_i$ for all $1 \leq i \leq j$. Then $\varphi(d) \mid \varphi(n)$, since

$$\frac{\varphi(n)}{\varphi(d)} = \prod_{i=1}^j \frac{p_i^{\alpha_i-1} (p_i - 1)}{p_i^{\beta_i-1} (p_i - 1)} \cdot \prod_{i=j+1}^k p_i^{\alpha_i-1} (p_i - 1) = \prod_{i=1}^j p_i^{\alpha_i-\beta_i} \cdot \prod_{i=j+1}^k p_i^{\alpha_i-1} (p_i - 1) \in \mathbb{Z}. \quad \blacksquare$$

0.3 $\mathbb{Z}/n\mathbb{Z}$: The Integers Modulo n **Exercise 0.3.1**

Write down explicitly all the elements in the residue classes of $\mathbb{Z}/18\mathbb{Z}$.

Solution. Note that $\bar{x} = \{x + 18k \mid k \in \mathbb{Z}\}$. Explicitly, we have:

$$\begin{array}{lll} \bar{0} = \{0, 18, -18, 36, -36, \dots\} & \bar{1} = \{1, 19, -17, 37, -35, \dots\} & \bar{2} = \{2, 20, -16, 38, -34, \dots\} \\ \bar{3} = \{3, 21, -15, 39, -33, \dots\} & \bar{4} = \{4, 22, -14, 40, -32, \dots\} & \bar{5} = \{5, 23, -13, 41, -31, \dots\} \\ \bar{6} = \{6, 24, -12, 42, -30, \dots\} & \bar{7} = \{7, 25, -11, 43, -29, \dots\} & \bar{8} = \{8, 26, -10, 44, -28, \dots\} \\ \bar{9} = \{9, 27, -9, 45, -27, \dots\} & \bar{10} = \{10, 28, -8, 46, -26, \dots\} & \bar{11} = \{11, 29, -7, 47, -25, \dots\} \\ \bar{12} = \{12, 30, -6, 48, -24, \dots\} & \bar{13} = \{13, 31, -5, 49, -23, \dots\} & \bar{14} = \{14, 32, -4, 50, -22, \dots\} \\ \bar{15} = \{15, 33, -3, 51, -21, \dots\} & \bar{16} = \{16, 34, -2, 52, -20, \dots\} & \bar{17} = \{17, 35, -1, 53, -19, \dots\} \quad \blacksquare \end{array}$$

Exercise 0.3.2

Prove that the distinct equivalence classes in $\mathbb{Z}/n\mathbb{Z}$ are precisely $\bar{0}, \bar{1}, \bar{2}, \dots, \overline{n-1}$.

HINT. Use the Division Algorithm.

Solution. Let $m \in \mathbb{Z}$. By the Division Algorithm, there exists unique $q, r \in \mathbb{Z}$ such that

$$m = nq + r, \quad 0 \leq r < n.$$

Then $m \equiv r \pmod{n}$, so $m \in \bar{r}$. By definition, $\bar{r}$ is one of $\bar{0}, \bar{1}, \dots, \overline{n-1}$. Thus, every equivalence class in $\mathbb{Z}/n\mathbb{Z}$ is one of $\bar{0}, \bar{1}, \dots, \overline{n-1}$. Moreover, they are all distinct. Otherwise, if $\bar{x} = \bar{y}$ for $0 \leq x, y < n$, then $x \equiv y \pmod{n}$, or $n \mid (x - y)$. However, since $-n < x - y < n$, we must have $x - y = 0$, or $x = y$. $\blacksquare$

Exercise 0.3.3

Prove that if

$$a = a_n 10^n + a_{n-1} 10^{n-1} + \dots + a_1 10 + a_0$$

is any positive integer, then

$$a \equiv (a_n + a_{n-1} + \dots + a_1 + a_0) \pmod{9}.$$

HINT. Use the fact that $10 \equiv 1 \pmod{9}$.

REMARK. Note that this is the usual arithmetic rule that the remainder after division by 9 is the same as the sum of the decimal digits mod 9—in particular an integer is divisible by 9 if and only if the sum of its digits is divisible by 9.

Solution. Using the hint, then

$$\begin{aligned} a &\equiv (a_n 10^n + a_{n-1} 10^{n-1} + \dots + a_1 10 + a_0) \pmod{9} \\ &\equiv (a_n 1^n + a_{n-1} 1^{n-1} + \dots + a_1 1 + a_0) \pmod{9} \\ &= (a_n + a_{n-1} + \dots + a_1 + a_0) \pmod{9} \quad \blacksquare \end{aligned}$$

Exercise 0.3.4

Compute the remainder when 37^{100} is divided by 29.

Solution. Modding 37^{100} by 29 results in 23. ■

Exercise 0.3.5

Compute the last two digits of 9^{1500} .

Solution. Modding 9^{1500} by 100 results in 1. ■

Exercise 0.3.6

Prove that the square of the elements in $\mathbb{Z}/4\mathbb{Z}$ are just $\bar{0}$ and $\bar{1}$.

Solution.

$$0^2 = 0 \equiv 0 \pmod{4}$$

$$1^2 = 1 \equiv 1 \pmod{4}$$

$$2^2 = 4 \equiv 0 \pmod{4}$$

$$3^2 = 9 \equiv 1 \pmod{4}$$
 ■

Exercise 0.3.7

Prove that for any integers a and b that $a^2 + b^2$ never leaves a remainder of 3 when divided by 4.

HINT. Use [Exercise 0.3.6](#).

Solution. By [Exercise 0.3.6](#), a^2 and b^2 are both either 0 mod 4 or 1 mod 4, so their sum can be:

$$0 + 0 \equiv 0 \pmod{4}$$

$$0 + 1 \equiv 1 \pmod{4}$$

$$1 + 0 \equiv 1 \pmod{4}$$

$$1 + 1 \equiv 2 \pmod{4}$$

In any of the sums, there is no remainder of 3 when dividing by 4. ■

Exercise 0.3.8

Prove that the equation $a^2 + b^2 = 3c^2$ has no solutions in nonzero integers a , b and c .

HINT. Consider the equation mod 4 as in the previous two exercises and show that a , b and c would all have to be divisible by 2. Then each of a^2 , b^2 and c^2 has a factor of 4 and by dividing through by 4 show that there would be a smaller set of solutions to the original equation. Iterate to reach a contradiction.

Solution. We consider $a^2 + b^2 = 3c^2$ in mod 4. By [Exercise 0.3.7](#), $a^2 + b^2$ is either 0, 1, or 2 mod 4. By [Exercise 0.3.6](#), c^2 is either 0 or 1 mod 4, so that $3c^2$ is either 0 or 3 mod 4. Thus, both sides must be 0 mod 4.

It is clear that $4 \mid 3c^2$ implies $4 \mid c^2$, and hence $2 \mid c$. Observe that if a and b were both odd, then $a^2 + b^2 \equiv 2 \pmod{4}$. If one of a or b were odd and the other even, then $a^2 + b^2 \equiv 1 \pmod{4}$. Thus, both a and b must be even. Put $a = 2\alpha$, $b = 2\beta$, and $c = 2\gamma$ for nonzero $\alpha, \beta, \gamma \in \mathbb{Z}$. Substituting into the original equation gives

$$(2\alpha)^2 + (2\beta)^2 = 3(2\gamma)^2.$$

We may divide by 4 to obtain $\alpha^2 + \beta^2 = 3\gamma^2$, which is the same form as the original equation but with integers with smaller absolute values. Repeating this process leads to an infinite descent, which is impossible. Therefore, there are no solutions in nonzero integers a, b , and c . ■

Exercise 0.3.9

Prove that the square of any odd integer always leaves a remainder of 1 when divided by 8.

Solution. Put $a = 2k + 1$ for $k \in \mathbb{Z}$. Then $a^2 = 4k(k + 1) + 1 \equiv 1 \pmod{8}$ since $k(k + 1)$ is always even. ■

Exercise 0.3.10

Prove that the number of elements of $(\mathbb{Z}/n\mathbb{Z})^\times$ is $\varphi(n)$, where φ denotes the Euler φ -function.

Solution. It suffices to show that $\bar{a} \in (\mathbb{Z}/n\mathbb{Z})^\times$ if and only if $(a, n) = 1$.

($\Rightarrow$) If $\bar{a} \in (\mathbb{Z}/n\mathbb{Z})^\times$, then there exists $\bar{b} \in \mathbb{Z}/n\mathbb{Z}$ such that $\bar{a} \cdot \bar{b} = \bar{1}$. Then $ab \equiv 1 \pmod{n}$ implies the existence of $k \in \mathbb{Z}$ such that $ab - 1 = kn$, or $ab - kn = 1$. Thus, $(a, n) = 1$.

($\Leftarrow$) By $(a, n) = 1$, there exists $x, y \in \mathbb{Z}$ such that $ax + ny = 1$, or $ax = 1 - ny$. We then have $ax \equiv 1 \pmod{n}$, so that $\bar{a} \cdot \bar{x} = \bar{1}$. Thus, $\bar{a} \in (\mathbb{Z}/n\mathbb{Z})^\times$. ■

Exercise 0.3.11

Prove that if $\bar{a}, \bar{b} \in (\mathbb{Z}/n\mathbb{Z})^\times$, then $\bar{a} \cdot \bar{b} \in (\mathbb{Z}/n\mathbb{Z})^\times$.

Solution. Since $\bar{a}, \bar{b} \in (\mathbb{Z}/n\mathbb{Z})^\times$, there exist $\bar{c}, \bar{d} \in \mathbb{Z}/n\mathbb{Z}$ such that $\bar{a} \cdot \bar{c} = \bar{b} \cdot \bar{d} = \bar{1}$. Then

$$(\bar{a} \cdot \bar{b})(\bar{c} \cdot \bar{d}) = (\bar{a} \cdot \bar{c})(\bar{b} \cdot \bar{d}) = \bar{1} \cdot \bar{1} = \bar{1},$$

so that $\bar{a} \cdot \bar{b} \in (\mathbb{Z}/n\mathbb{Z})^\times$. ■

Exercise 0.3.12

Let $n \in \mathbb{Z}$, $n > 1$ and let $a \in \mathbb{Z}$ with $1 \leq a \leq n$. Prove if a and n are not relatively prime, there exists an integer b with $1 \leq b < n$ such that $ab \equiv 0 \pmod{n}$ and deduce that there cannot be an integer c such that $ac \equiv 1 \pmod{n}$.

Solution. Let $(a, n) = d > 1$. Then $a/d, n/d \in \mathbb{Z}$ and $1 \leq a/d < n/d$. Put $b = n/d$, so that $1 \leq b < n$ and

$$ab = a(n/d) = (a/d)n \equiv 0 \pmod{n}.$$

If there existed some $c \in \mathbb{Z}$ such that $ac \equiv 1 \pmod{n}$, then $(ac)b \equiv b \pmod{n}$. However, $ab \equiv 0 \pmod{n}$, so that $0 \equiv b \pmod{n}$, which is impossible since $1 \leq b < n$. ■

Exercise 0.3.13

Let $n \in \mathbb{Z}, n > 1$ and let $a \in \mathbb{Z}$ with $1 \leq a \leq n$. Prove that if a and n are relatively prime then there is an integer c such that $ac \equiv 1 \pmod{n}$.

HINT. Use the fact that the GCD of two integers is a $\mathbb{Z}$ -linear combination of the integers.

Solution. Since $(a, n) = 1$, there exists $c, x \in \mathbb{Z}$ such that $ac + nx = 1$. Then $ac \equiv 1 \pmod{n}$. ■

Exercise 0.3.14

Conclude from the previous two exercises that $(\mathbb{Z}/n\mathbb{Z})^\times$ is the set of elements $\bar{a}$ of $\mathbb{Z}/n\mathbb{Z}$ with $(a, n) = 1$ and hence prove Proposition 4. Verify this directly in the case $n = 12$.

Solution. The previous two exercises show that a and n are relatively prime if and only if there exists b such that $ab \equiv 1 \pmod{n}$, which is exactly the proposition. For $n = 12$, the elements 1, 5, 7, 11 are relatively prime to 12, so that $(\mathbb{Z}/12\mathbb{Z})^\times = \{\bar{1}, \bar{5}, \bar{7}, \bar{11}\}$ whose inverses are $\bar{1}, \bar{5}, \bar{5}, \bar{11}$ respectively. ■

Exercise 0.3.15

For each of the following pairs of integers a and n , show that a is relatively prime to n and determine the multiplicative inverse of $\bar{a}$ in $\mathbb{Z}/n\mathbb{Z}$.

- (a) $a = 13, n = 20$
- (b) $a = 69, n = 89$
- (c) $a = 1891, n = 3797$
- (d) $a = 6003722857, n = 77695236973$

Solution. Use the Euclidean Algorithm to find $x, y \in \mathbb{Z}$ such that $ax + ny = 1$ for each pair (a, n) . Then $ax \equiv 1 \pmod{n}$, and x is the multiplicative inverse of $\bar{a}$ in $\mathbb{Z}/n\mathbb{Z}$.

- (a) 17.
- (b) 40.
- (c) 253.
- (d) 77695236753.

Exercise 0.3.16

Write a computer program to add and multiply mod n , for any n given as input. The output of these operations should be the least residues of the sums and products of two integers. Also include the feature that if $(a, n) = 1$, an integer c between 1 and $n - 1$ such that $\bar{a} \cdot \bar{c} = \bar{1}$ may be printed on request.

REMARK. Your program should not, of course, simply quote “mod” functions already built into many systems.

Solution. Please see `modular_arithmetic.py` for the solution. ■

1 Introduction to Groups

1.1 Basic Axioms and Examples

Let G be a group.

Exercise 1.1.1

Determine which of the following binary operations are associative.

- (a) The operation $\star$ on $\mathbb{Z}$ defined by $a \star b = a - b$.
- (b) The operation $\star$ on $\mathbb{R}$ defined by $a \star b = a + b + ab$.
- (c) The operation $\star$ on $\mathbb{Q}$ defined by

$$a \star b = \frac{a+b}{5}.$$

- (d) The operation $\star$ on $\mathbb{Z} \times \mathbb{Z}$ defined by $(a, b) \star (c, d) = (ad + bc, bd)$.
- (e) The operation $\star$ on $\mathbb{Q} - \{0\}$ defined by

$$a \star b = \frac{a}{b}.$$

Solution.

- (a) Not associative: $(1 \star 0) \star 1 = 0 \neq 2 = 1 \star (0 \star 1)$.
- (b) Associative.
- (c) Not associative: $(1 \star 0) \star 2 = 11/25 \neq 7/25 = 1 \star (0 \star 2)$.
- (d) Associative.
- (e) Not associative: $(1 \star 2) \star 3 = 1/6 \neq 3/2 = 1 \star (2 \star 3)$. ■

Exercise 1.1.2

Decide which of the binary operations in the preceding exercise are commutative.

- (a) The operation $\star$ on $\mathbb{Z}$ defined by $a \star b = a - b$.
- (b) The operation $\star$ on $\mathbb{R}$ defined by $a \star b = a + b + ab$.
- (c) The operation $\star$ on $\mathbb{Q}$ defined by

$$a \star b = \frac{a+b}{5}.$$

- (d) The operation $\star$ on $\mathbb{Z} \times \mathbb{Z}$ defined by $(a, b) \star (c, d) = (ad + bc, bd)$.
- (e) The operation $\star$ on $\mathbb{Q} - \{0\}$ defined by

$$a \star b = \frac{a}{b}.$$

Solution.

- (a) Not commutative: $1 - 0 \neq 0 - 1$.
- (b) Commutative.
- (c) Commutative.
- (d) Commutative.
- (e) Not commutative: $1/2 \neq 2/1$. ■

Exercise 1.1.3

Prove that addition of residue classes in $\mathbb{Z}/n\mathbb{Z}$ is associative (you may assume it is well-defined).

Solution. Let $\bar{a}, \bar{b}, \bar{c} \in \mathbb{Z}/n\mathbb{Z}$. By associativity of addition in $\mathbb{Z}$, we have:

$$(\bar{a} + \bar{b}) + \bar{c} = \overline{(a + b) + c} = \overline{a + (b + c)} = \bar{a} + (\bar{b} + \bar{c}).$$

Exercise 1.1.4

Prove that multiplication of residue classes in $\mathbb{Z}/n\mathbb{Z}$ is associative (you may assume it is well-defined).

Solution. Use the same argument as in the previous exercise, replacing $+$ with $\cdot$.

Exercise 1.1.5

Prove for all $n > 1$ that $\mathbb{Z}/n\mathbb{Z}$ is not a group under multiplication of residue classes.

Solution. Since $0 \cdot a = 0$ for all $a \in \mathbb{Z}$, so $\bar{a} \cdot \bar{0} = \bar{0}$.

Exercise 1.1.6

Determine which of the following sets are groups under addition:

- (a) the set of rational numbers (including $0 = 0/1$) in lowest terms whose denominators are odd
- (b) the set of rational numbers (including $0 = 0/1$) in lowest terms whose denominators are even
- (c) the set of rational numbers of absolute value < 1
- (d) the set of rational numbers of absolute value ≥ 1 together with 0
- (e) the set of rational numbers with denominators equal to 1 or 2
- (f) the set of rational numbers with denominators equal to 1, 2, or 3

Solution. Let S denote the set in each part.

- (a) S has identity element 0, inverse element $-a/b$ for $a/b \in S$, and inherits associativity from $\mathbb{Q}$. Moreover, if $a/b, c/d \in S$ such that $(a, b) = (c, d) = 1$, then

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd},$$

where bd is odd. Since an odd number cannot have a factor of 2, the denominator remains odd when reduced to lowest terms. Hence, S is closed under addition, and it is a group.

- (b) No: $1/2 \in S$, but $1/2 + 1/2 = 2/2 = 1/1 \notin S$.

- (c) Same as (b).

- (d) No: $3/2, 1 \in S$, but $3/2 - 1 = 1/2 \notin S$.

- (e) Identity, inverses, and associativity are satisfied. Let $a, b \in \mathbb{Z}$. Then

$$\frac{a}{1} + \frac{b}{1} = \frac{a+b}{1}, \quad \frac{a}{2} + \frac{b}{2} = \frac{a+b}{2}, \quad \frac{a}{1} + \frac{b}{2} = \frac{2a+b}{2},$$

where the last case is chosen without loss of generality. Hence, S is closed.

- (f) No: $1/3, 1/2 \in S$, but $1/3 + 1/2 = 5/6 \notin S$.

Exercise 1.1.7

Let $G = \{x \in \mathbb{R} \mid 0 \leq x < 1\}$ and for $x, y \in G$ let $x \star y$ be the fractional part of $x + y$ (i.e., $x \star y = x + y - [x + y]$ where $[a]$ is the greatest integer less than or equal to a). Prove that $\star$ is a well-defined binary operation on G and that G is an Abelian group under $\star$.

Solution. Let $x, y, z \in G$.

- Note that $0 \leq x + y < 2$. If $x + y < 1$, then $[x + y] = 0$, and $x \star y = x + y \in G$. If $1 \leq x + y < 2$, then $[x + y] = 1$, and $x \star y = x + y - 1 \in G$. Hence, $x \star y \in G$, and $\star$ is well-defined.
- $x \star 0 = x - [x] = x$. Then 0 is the identity element of G under $\star$.
- $x \star (1 - x) = x + (1 - x) - [x + (1 - x)] = 0$. Hence, $1 - x$ is the inverse of x under $\star$.
- $x \star y = x + y - [x + y] = y + x - [y + x] = y \star x$. Hence, G is Abelian under $\star$.
- Note that $[\alpha + \beta] = [\alpha] + \beta$ for $\beta \in \mathbb{Z}$. Then

$$\begin{aligned}
 (x \star y) \star z &= (x + y - [x + y]) \star z \\
 &= (x + y + z - [x + y]) - [x + y + z - [x + y]] \\
 &= x + y + z - [x + y] - [x + y + z - [x + y]] \\
 &= x + y + z - [x + y] - ([x + y + z] - [x + y]) \\
 &= x + y + z - [x + y + z] \\
 &= x \star (y + z - [y + z]) = x \star (y \star z).
 \end{aligned}$$

■

Exercise 1.1.8

Let $G = \{z \in \mathbb{C} \mid z^n = 1 \text{ for some } n \in \mathbb{Z}^+\}$.

- Prove that G is a group under multiplication (called the **group of roots of unity in $\mathbb{C}$**).
- Prove that G is not a group under addition.

Solution.

- Since $1^1 = 1$, then $1 \in G$. If $z \in G$ for some n , then $(z^{-1})^n = (z^n)^{-1} = 1^{-1} = 1$, so that $z^{-1} \in G$. Associativity from $\mathbb{C}$ is inherited by G . Finally, let $z, w \in G$ such that $z^n = w^m = 1$ for $n, m \in \mathbb{Z}^+$. Then

$$(zw)^{nm} = z^{nm} w^{nm} = (z^n)^m (w^m)^n = 1 \cdot 1 = 1,$$

so that $zw \in G$. Hence, G is a group under multiplication.

- No; note $1, -1 \in G$, but $1 + (-1) = 0 \notin G$ since $0^n = 0 \neq 1$ for any $n \in \mathbb{Z}^+$.

■

Exercise 1.1.9

Let $G = \{a + b\sqrt{2} \in \mathbb{R} \mid a, b \in \mathbb{Q}\}$.

- Prove that G is a group under addition.
- Prove that the nonzero elements of G are a group under multiplication.

HINT. “Rationalize the denominators” to find multiplicative inverses.

Solution.

- Note that addition of two elements in G is given by

$$(a + b\sqrt{2}) + (c + d\sqrt{2}) = (a + c) + (b + d)\sqrt{2} \in G.$$

From this, we see that G is closed under addition. The identity is $0 + 0\sqrt{2} = 0$, and the inverse of $a + b\sqrt{2}$ is $-a - b\sqrt{2}$. Associativity is inherited from $\mathbb{R}$. Hence, G is a group under addition.

(b) Note that multiplication of two elements in G is given by

$$(a + b\sqrt{2})(c + d\sqrt{2}) = (ac + 2bd) + (ad + bc)\sqrt{2} \in G.$$

From this, we see that G is closed under multiplication. The identity is $1 + 0\sqrt{2} = 1$, and the inverse of a nonzero element $a + b\sqrt{2}$ is

$$(a + b\sqrt{2})^{-1} = \frac{1}{a + b\sqrt{2}} = \frac{1}{a + b\sqrt{2}} \cdot \frac{a - b\sqrt{2}}{a - b\sqrt{2}} = \frac{a - b\sqrt{2}}{a^2 - 2b^2} = \frac{a}{a^2 - 2b^2} - \frac{b}{a^2 - 2b^2}\sqrt{2}.$$

Observe that $a - b\sqrt{2} \neq 0$, since that would imply $a = b\sqrt{2}$, contradicting that $a \in \mathbb{Q}$. Moreover, $a + b\sqrt{2} \neq 0$ implies $a^2 - 2b^2 \neq 0$. Hence, the inverse is well-defined and in G . Associativity is inherited from $\mathbb{R}$. Hence, the nonzero elements of G are a group under multiplication. ■

Exercise 1.1.10

Prove that a finite group is Abelian if and only if its group table is a symmetric matrix.

Solution. Let $G = \{g_1, g_2, \dots, g_n\}$ be a finite group.

($\Rightarrow$) If G is Abelian, then $g_i g_j = g_j g_i$ for all i, j . Hence, the (i, j) -th entry of the group table is equal to the (j, i) -th entry for all i, j , so that the group table is symmetric.

($\Leftarrow$) If the group table is symmetric, then $g_i g_j = g_j g_i$ for all i, j . Hence, G is Abelian. ■

Exercise 1.1.11

Find the orders of each element of the additive group $\mathbb{Z}/12\mathbb{Z}$.

Solution. For each $\bar{x} \in \mathbb{Z}/12\mathbb{Z}$, add it to itself until we arrive at $\bar{0}$. $|\bar{0}| = 1$, and $\bar{1} = 12$ since $12 \cdot \bar{1} = \bar{12} = \bar{0}$. In particular, we have:

$\bar{x}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{6}$	$\bar{7}$	$\bar{8}$	$\bar{9}$	$\bar{10}$	$\bar{11}$
$ \bar{x} $	1	12	6	4	3	12	2	12	3	4	6	12

Exercise 1.1.12

Find the orders of the following elements of the multiplicative group $(\mathbb{Z}/12\mathbb{Z})^\times$: $\bar{1}, \bar{-1}, \bar{5}, \bar{7}, \bar{-7}, \bar{13}$.

Solution.

$\bar{x}$	$\bar{1}$	$\bar{-1}$	$\bar{5}$	$\bar{7}$	$\bar{-7}$	$\bar{13}$
$ \bar{x} $	1	2	2	2	2	1

Exercise 1.1.13

Find the orders of the following elements of the additive group $\mathbb{Z}/36\mathbb{Z}$: $\bar{1}, \bar{2}, \bar{6}, \bar{9}, \bar{10}, \bar{12}, \bar{-1}, \bar{-10}, \bar{-18}$.

Solution.

$\bar{x}$	$\bar{1}$	$\bar{2}$	$\bar{6}$	$\bar{9}$	$\bar{10}$	$\bar{12}$	$\bar{-1}$	$\bar{-10}$	$\bar{-18}$
$ \bar{x} $	36	18	6	4	18	3	36	18	2

Exercise 1.1.14

Find the orders of the following elements of the multiplicative group $(\mathbb{Z}/36\mathbb{Z})^\times$: $\bar{1}, \bar{-1}, \bar{5}, \bar{13}, \bar{-13}, \bar{17}$.

Solution.

$\bar{x}$	$\bar{1}$	$\bar{-1}$	$\bar{5}$	$\bar{13}$	$\bar{-13}$	$\bar{17}$
$ \bar{x} $	1	2	6	3	6	2

■

Exercise 1.1.15

Prove that $(a_1 a_2 \dots a_n)^{-1} = a_n^{-1} a_{n-1}^{-1} \dots a_1^{-1}$ for all $a_1, a_2, \dots, a_n \in G$.

Solution. We proceed by induction. This holds for $n = 2$ by Proposition 1.1 (iv). Suppose it holds for some $n \in \mathbb{Z}^+$. Then for $n + 1$, we have

$$(a_1 a_2 \dots a_n a_{n+1})^{-1} = a_{n+1}^{-1} (a_1 a_2 \dots a_n)^{-1} = a_{n+1}^{-1} a_n^{-1} a_{n-1}^{-1} \dots a_1^{-1}.$$

■

Exercise 1.1.16

Let x be an element of G . Prove that $x^2 = 1$ if and only if $|x|$ is either 1 or 2.

Solution.

($\Rightarrow$) Suppose $x^2 = 1$. Then $|x| \leq 2$. Since $|x| \in \mathbb{Z}^+$, then $|x|$ is either 1 or 2.

($\Leftarrow$) If $|x| = 1$, then $x = 1$ and $x^2 = 1$. If $|x| = 2$, then $x^2 = 1$ by definition.

■

Exercise 1.1.17

Let x be an element of G . Prove that if $|x| = n$ for some positive integer n , then $x^{-1} = x^{n-1}$.

Solution. Note that $x^n = x^{n-1}x = 1$. Then $x^{n-1} = x^{-1}$.

■

Exercise 1.1.18

Let x, y be elements of G . Prove that the following are equivalent:

(i) $xy = yx$; (ii) $y^{-1}xy = x$; (iii) $x^{-1}y^{-1}xy = 1$.

Solution. $xy = yx \iff y^{-1}xy = y^{-1}yx = x \iff x^{-1}y^{-1}xy = x^{-1}x = 1$.

■

Exercise 1.1.19

Let $x \in G$ and let $a, b \in \mathbb{Z}^+$.

(a) Prove that $x^{a+b} = x^a x^b$ and $(x^a)^b = x^{ab}$.

(b) Prove that $(x^a)^{-1} = x^{-a}$.

(c) Establish part (a) for arbitrary integers a and b .

Solution.

(a) There are a instances of x in x^a and b instances of x in x^b . Hence, there are $a + b$ instances of x in $x^a x^b$, so that $x^{a+b} = x^a x^b$. There are a instances of x in x^a , and there are b instances of x^a in $(x^a)^b$. Hence, there are ab instances of x in $(x^a)^b$, so that $(x^a)^b = x^{ab}$.

(b) By [Exercise 1.1.15](#), then $(x^a)^{-1}$ consists of a instances of x^{-1} , so that $(x^a)^{-1} = x^{-a}$.

(c) We consider the following cases:

- If $a, b \in \mathbb{Z}^+$, then the result follows from part (a).
- If $a \in \mathbb{Z}^+$ and $b < 0$ (the case of $a < 0$ and $b \in \mathbb{Z}^+$ is analogous), suppose $a + b \geq 0$. Then

$$x^a = x^{a+b-b} = x^{a+b}x^{-b},$$

so that $x^{a+b} = x^a x^b$. If $a + b < 0$, then

$$(x^{a+b})^{-1} = x^{-a-b} = x^{-b}x^{-a} = (x^a x^b)^{-1},$$

implying that $x^{a+b} = x^a x^b$, since inverses are unique. Moreover,

$$(x^a)^b = (x^a)^{-(-b)} = ((x^a)^{-b})^{-1} = (x^{-ab})^{-1} = x^{ab}.$$

- If $a < 0$ and $b < 0$, then $a + b < 0$ and

$$x^{a+b} = x^{b+a} = (x^{-b-a})^{-1} = (x^{-b}x^{-a})^{-1} = (x^{-a})^{-1}(x^{-b})^{-1} = x^a x^b.$$

Moreover,

$$(x^a)^b = ((x^{-a})^{-1})^b = ((x^{-a})^b)^{-1} = (x^{(-a)b})^{-1} = x^{ab}. \quad \blacksquare$$

Exercise 1.1.20

For x an element in G , show that x and x^{-1} have the same order.

Solution. Suppose $|x| = n \in \mathbb{Z}^+$. Then

$$(x^{-1})^n = (x^n)^{-1} = 1^{-1} = 1$$

so that $|x^{-1}| \leq n$ and has finite order. Similarly, suppose $|x^{-1}| = m \in \mathbb{Z}^+$. Then

$$x^m = ((x^{-1})^m)^{-1} = 1^{-1} = 1$$

so that $|x| \leq m$ and has finite order. Then $n = m$, and $|x| = |x^{-1}|$. Specifically, x has finite order if and only if x^{-1} has finite order, hence x has infinite order if and only if x^{-1} has infinite order. $\blacksquare$

Exercise 1.1.21

Let G be a finite group and let x be an element of order n . Prove that if n is odd, then $x = (x^2)^k$ for some $k \geq 1$.

Solution. Put $n = 2k - 1$ for $k \in \mathbb{Z}^+$. Then $x^n = x^{2k-1} = 1$, so that $x^{2k} = x$. Hence, $x = (x^2)^k$. $\blacksquare$

Exercise 1.1.22

If x and g are elements of the group G , prove that $|x| = |g^{-1}xg|$. Deduce that $|ab| = |ba|$ for all $a, b \in G$.

Solution. We first prove the following lemma.

Lemma: $(g^{-1}xg)^n = g^{-1}x^ng$ for all $n \in \mathbb{Z}^+$.

Proof. We proceed by induction, where $n = 1$ is clear. If it holds for $n \in \mathbb{Z}^+$, then for $n + 1$, we have

$$(g^{-1}xg)^{n+1} = (g^{-1}xg)^n(g^{-1}xg) = (g^{-1}x^ng)(g^{-1}xg) = g^{-1}x^{n+1}g.$$

Hence, the result follows by induction for all $n \in \mathbb{Z}^+$. ■

Let $|x| = m \in \mathbb{Z}^+$. Then

$$(g^{-1}xg)^m = g^{-1}x^mg = g^{-1}1g = 1,$$

so that $|g^{-1}xg| \leq m$. Similarly, let $|g^{-1}xg| = n \in \mathbb{Z}^+$. Then

$$x^n = (gg^{-1})x^n(gg^{-1}) = g(g^{-1}xg)^ng^{-1} = g1g^{-1} = 1,$$

so that $|x| \leq n$. Hence, $m = n$, and $|x| = |g^{-1}xg|$.

Let $a, b \in G$. Then $|ab| = |b^{-1}(ab)b| = |ba|$ by the above result. ■

Exercise 1.1.23

Suppose $x \in G$ and $|x| = n < \infty$. If $n = st$ for some positive integers s and t , prove that $|x^s| = t$.

Solution. Let $|x^s| = r \in \mathbb{Z}^+$. Then

$$(x^s)^t = x^{st} = x^n = 1,$$

so that $r \leq t$. Moreover,

$$x^{sr} = (x^s)^r = 1$$

implies that $|x| = st \leq sr$, hence $t \leq r$. Thus, $|x^s| = r = t$. ■

Exercise 1.1.24

If a and b are commuting elements of G , prove that $(ab)^n = a^n b^n$ for all $n \in \mathbb{Z}$.

HINT. Do this by induction for positive n first.

Solution. The case for $n = 0$ and $n = 1$ is clear. If it holds for some $n \in \mathbb{Z}^+$, then for $n + 1$, we have

$$(ab)^{n+1} = (ab)^n(ab) = (a^n b^n)(ab) = a^n (b^n a)b = a^n (ab^n)b = a^{n+1} b^{n+1}.$$

For $n < 0$, then $(ab)^n = ((ab)^{-n})^{-1} = (b^{-n} a^{-n})^{-1} = a^n b^n$. ■

Exercise 1.1.25

Prove that if $x^2 = 1$ for all $x \in G$, then G is Abelian.

Solution. Since $x^2 = 1$, then $x = x^{-1}$ for all $x \in G$. Let $a, b \in G$. Then

$$ab = (ab)^{-1} = b^{-1} a^{-1} = ba. \quad \blacksquare$$

Exercise 1.1.26

Assume H is a nonempty subset of $(G, \star)$ which is closed under the binary operation on G and is closed under inverses, i.e., for all $h, k \in H$, $hk, h^{-1} \in H$. Prove that H is a group under the operation $\star$ restricted to H (such a subset H is called a **subgroup** of G).

Solution. Let $H \subseteq (G, \star)$ be nonempty and closed under the binary operation on G and inverses. Let $h \in H$. Then $h^{-1} \in H$, so that $hh^{-1} = 1 \in H$. Hence, H has an identity. Associativity is inherited from G . Hence, $(H, \star|_H)$ is a group. ■

Exercise 1.1.27

Prove that if x is an element of the group G , then $\{x^n \mid n \in \mathbb{Z}\}$ is a **subgroup** of G (called the **cyclic subgroup** of G generated by x).

Solution. Let H be the set in question. Then $1 = x^0 \in H$, so that H is nonempty. Let $x^m, x^n \in H$. Then $x^m x^n = x^{m+n} \in H$ and $(x^m)^{-1} = x^{-m} \in H$. Hence, H is closed under the binary operation on G and inverses. By Exercise 1.1.26, then H is a subgroup of G . ■

Exercise 1.1.28

Let $(A, \star)$ and $(B, \diamond)$ be groups and let $A \times B$ be their direct product (as defined in Example 6). Verify all the group axioms for $A \times B$:

(a) Prove that the associative law holds:

$$\text{for all } (a_1, b_1), (a_2, b_2), (a_3, b_3) \in A \times B, [(a_1, b_1)(a_2, b_2)](a_3, b_3) = (a_1, b_1)[(a_2, b_2)(a_3, b_3)]$$

(a) Prove that $(1, 1)$ is the identity of $A \times B$, and

(b) Prove that the inverse of (a, b) is (a^{-1}, b^{-1}) .

Solution.

(a) Let $(a_1, b_1), (a_2, b_2), (a_3, b_3) \in A \times B$. Then

$$\begin{aligned} [(a_1, b_1)(a_2, b_2)](a_3, b_3) &= (a_1 a_2, b_1 b_2)(a_3, b_3) \\ &= ((a_1 a_2) a_3, (b_1 b_2) b_3) \\ &= (a_1 (a_2 a_3), b_1 (b_2 b_3)) \\ &= (a_1, b_1)(a_2 a_3, b_2 b_3) \\ &= (a_1, b_1)[(a_2, b_2)(a_3, b_3)] \end{aligned}$$

(b) For $a \in A, b \in B$, we have $(a, b)(1, 1) = (a \star 1, b \diamond 1) = (a, b)$.

(c) $(a, b)(a^{-1}, b^{-1}) = (a \star a^{-1}, b \diamond b^{-1}) = (1, 1)$. ■

Exercise 1.1.29

Prove that $A \times B$ is an Abelian group if and only if both A and B are Abelian.

Solution. Let $a, a' \in A$ and $b, b' \in B$. Suppose $A \times B$ is Abelian. Then

$$(aa', bb') = (a, b)(a', b') = (a', b')(a, b) = (a'a, b'b)$$

so that A and B are Abelian. If A and B are both Abelian, then

$$(a, b)(a', b') = (aa', bb') = (a'a, b'b) = (a', b')(a, b)$$

hence $A \times B$ is Abelian. ■

Exercise 1.1.30

Prove that the elements $(a, 1)$ and $(1, b)$ of $A \times B$ commute and deduce that the order of (a, b) is the least common multiple of $|a|$ and $|b|$.

Solution. Let $a \in A$ and $b \in B$. Then

$$(a, 1)(1, b) = (a \cdot 1, 1 \cdot b) = (a, b) = (1 \cdot a, b \cdot 1) = (1, b)(a, 1)$$

so that $(a, 1)$ and $(1, b)$ commute. Let $\ell = \text{lcm}(|a|, |b|)$. Then

$$(a, b)^\ell = (a^\ell, b^\ell) = (1, 1)$$

so that $|(a, b)| \leq \ell$. Moreover, if $(a, b)^k = (1, 1)$ for some $k \in \mathbb{Z}^+$, then $a^k = 1$ and $b^k = 1$ so that both $|a|$ and $|b|$ divide k . Hence, ℓ divides k so that $\ell \leq k$. Thus, $|(a, b)| = \ell$. ■

Exercise 1.1.31

Prove that any finite group G of even order contains an element of order 2.

HINT. Let $t(G)$ be the set $\{g \in G \mid g \neq g^{-1}\}$. Show that $t(G)$ has an even number of elements and every non-identity element of $G - t(G)$ has order 2.

Solution. Let $g \in G$ with $g \neq g^{-1}$, so $g \in t(G)$. Since $(g^{-1})^{-1} = g$, then $g^{-1} \in t(G)$. Then elements of $t(G)$ come in pairs, so that $t(G)$ has even order. Since G has even order, then $G - t(G)$ has even order. Moreover, $1 \notin t(G)$ since $1^{-1} = 1$, hence G contains a non-identity element h . Then $h = h^{-1}$, so that $h^2 = 1$ and $|h| = 2$. ■

Exercise 1.1.32

If x is an element of finite order n in G , prove that the elements $1, x, x^2, \dots, x^{n-1}$ are all distinct. Deduce that $|x| \leq |G|$.

Solution. Let $a, b \in \mathbb{Z}^+$ be such that $0 \leq a < b < n$. Suppose $x^a = x^b$. Then

$$1 = x^b x^{-a} = x^{b-a}$$

so that $|x| \leq b - a < n$, which is a contradiction. Hence, the elements $1, x, x^2, \dots, x^{n-1}$ are all distinct. Since these elements are all in G , then $|x| = n \leq |G|$. ■

Exercise 1.1.33

Let x be an element of finite order n in G .

- (a) Prove that if n is odd then $x^i \neq x^{-i}$ for all $i = 1, 2, \dots, n-1$.
- (b) Prove that if $n = 2k$ and $1 \leq i < n$ then $x^i = x^{-i}$ if and only if $i = k$.

Solution.

- (a) Note that $x^{n-i} = x^{-i}$. If $i \neq n-i$, then $x^i \neq x^{n-i}$ by **Exercise 1.1.32**, so $x^i \neq x^{-i}$. If $i = n-i$, then $2i = n$, which is impossible since n is odd. Hence, $x^i \neq x^{-i}$ for all $i = 1, 2, \dots, n-1$.
- (b) Part (a) shows that $x^i \neq x^{-i}$ for $i \neq k$. If $i = k$, the result follows. ■

Exercise 1.1.34

If x is an element of infinite order in G , prove that the elements $x^n, n \in \mathbb{Z}$ are all distinct.

Solution. If $x^a = x^b$ for $a, b \in \mathbb{Z}$ such that $a < b$, then $x^{b-a} = 1$, contradicting that x has infinite order. Hence, the elements $x^n, n \in \mathbb{Z}$ are all distinct. ■

Exercise 1.1.35

If x is an element of finite order n in G , use the Division Algorithm to show that any integral power of x equals one of the elements in the set $\{1, x, x^2, \dots, x^{n-1}\}$.

REMARK. These are the distinct elements of the cyclic subgroup of G generated by x .

Solution. Let $|x| = n \in \mathbb{Z}^+$ and let $m \in \mathbb{Z}$. By the Division Algorithm, there exist unique integers q and r such that

$$m = qn + r$$

where $0 \leq r < n$. Then

$$x^m = x^{qn+r} = x^{qn}x^r = (x^n)^q x^r = 1^q x^r = x^r$$

so that any integral power of x equals one of the elements in the set $\{1, x, x^2, \dots, x^{n-1}\}$. ■

Exercise 1.1.36

Assume $G = \{1, a, b, c\}$ is a group of order 4 with identity 1. Assume also that G has no elements of order 4 (so by Exercise 1.1.32, every element has order ≤ 3). Use the cancellation laws to show that there is a unique group table for G . Deduce that G is Abelian.

Solution. Since G has even order, it has an element of order 2 by Exercise 1.1.31. Without loss of generality, let a be that element so that $a^2 = 1$. Since there are no elements of order 4, then the remaining elements b and c may have order 2 or 3.

We now observe that $ab \neq 1$ since that would imply that $b = a^{-1} = a$ contradicting that a and b are distinct elements. Similarly, $ab \neq a$ since that would imply that $b = 1$, and $ab \neq b$ since that would imply that $a = 1$. Thus, $ab = ba = c$. Similarly, $ac \neq 1, a, c$ so that $ac = ca = b$, hence $b^2 = (ca)(ac) = c^2$.

We now consider the possible orders of b and c . If $b^2 \neq 1$, then $|b| = 3$. But then $b^3 = bb^2 = bc = a$, contradicting that $|b| = 3$. Thus, $b^2 = 1$. Similarly, $c^2 = 1$. The group table for G is therefore as follows:

★	1	a	b	c
1	1	a	b	c
a	a	1	c	b
b	b	c	1	a
c	c	b	a	1

from which we may deduce that G is Abelian. ■

1.2 Dihedral Groups

In these exercises, D_{2n} has the usual presentation $D_{2n} = \langle r, s \mid r^n = s^2 = 1, rs = sr^{-1} \rangle$.

Exercise 1.2.1

Compute the order of each of the elements in the following groups:

- (a) D_6 (b) D_8 (c) D_{10}

Solution. For any reflection $sr^k \in D_{2n}$, it is clear that $sr^k \neq 1$. Moreover,

$$(sr^k)^2 = s(r^k s)r^k = s(sr^{-k})r^k = r^{-k}r^k = 1$$

so that every reflection has order 2. We now compute the orders of the rotations in each group.

(a) $D_6 = \{1, r, r^2, s, sr, sr^2\}$. Then $|1| = 1$ and $|r| = |r^2| = 3$.

(b) $D_8 = \{1, r, r^2, r^3, s, sr, sr^2, sr^3\}$. Then $|1| = 1$, $|r^2| = 2$ and $|r| = |r^3| = 4$.

(c) $D_{10} = \{1, r, r^2, r^3, r^4, s, sr, sr^2, sr^3, sr^4\}$. Then $|1| = 1$ and $|r| = |r^2| = |r^3| = |r^4| = 5$. ■

Exercise 1.2.2

Use the generators and relations above to show that if x is any element of D_{2n} which is not a power of r , then $rx = xr^{-1}$.

Solution. If x is not a power of r , then x is a reflection, i.e., $x = sr^k$ for some $0 \leq k < n$. Then

$$rx = r(sr^k) = (rs)r^k = (sr^{-1})r^k = s(r^{-1}r^k) = sr^{k-1} = (sr^k)r^{-1} = xr^{-1} \quad \blacksquare$$

Exercise 1.2.3

Use the generators and relations above to show that every element of D_{2n} which is not a power of r has order 2. Deduce that D_{2n} is generated by the two elements s and sr , both of which have order 2.

HINT. cf. [Exercise 1.1.33](#).

Solution. If an element is not a power of r , then it is a reflection. See [Exercise 1.2.1](#) as to why every reflection has order 2. Moreover, for any $r^k, sr^k \in D_{2n}$ for $0 \leq k < n$, we may rewrite them as

$$r^k = (s(sr))^k \quad \text{and} \quad sr^k = s(s(sr))^k,$$

showing that every element of D_{2n} can be expressed in terms of s and sr . ■

Exercise 1.2.4

If $n = 2k$ is even and $n \geq 4$, show that $z = r^k$ is an element of order 2 which commutes with all elements of D_{2n} . Show also that z is the only non-identity element of D_{2n} which commutes with all elements of D_{2n} .

HINT. cf. [Exercise 1.1.33](#).

Solution. Since $k \neq 1$, then $r^k \neq 1$. Moreover, $(r^k)^2 = r^n = 1$, hence $|r^k| = 2$. Since r^k commutes with other powers of r , consider $sr^j \in D_{2n}$ for $0 \leq j < n$. Then

$$r^k(sr^j) = (r^k s)r^j = (sr^{-k})r^j = s(r^{-k}r^j) = sr^{j-k} = (sr^j)r^k,$$

Now let $w \in D_{2n}$ such that $wx = xw$ for all $x \in D_{2n}$. If $w = r^m$ for $0 \leq m < n$, then commuting with s gives $w = w^{-1}$, which implies $m = k$ by [Exercise 1.1.33](#). If $w = sr^m$ instead, then commuting with r implies $r = r^{-1}$, which cannot be true since $n \geq 4$. ■

Exercise 1.2.5

If n is odd and $n \geq 3$, show that the identity is the only element of D_{2n} which commutes with all elements of D_{2n} .

HINT. cf. [Exercise 1.1.33](#).

Solution. Using the same argument as the previous exercise and the odd case in [Exercise 1.1.33](#), then w cannot be a rotation since $r^k \neq r^{-k}$. The reflection case is also the same as the previous exercise. ■

Exercise 1.2.6

Let x and y be elements of order 2 in any group G . Prove that if $t = xy$, then $tx = xt^{-1}$ (so that if $n = |xy| < \infty$ then x, y satisfy the same relations in G as s, r do in D_{2n}).

Solution. Since $|x| = |y| = 2$, then $x = x^{-1}$ and $y = y^{-1}$. Then

$$tx = (xy)x = x(yx) = x(xy)^{-1} = xt^{-1}$$

Exercise 1.2.7

Show that $\langle a, b \mid a^2 = b^2 = (ab)^n = 1 \rangle$ gives a presentation for D_{2n} in terms of the two generators a and b of order 2 computed in [Exercise 1.2.3](#) above.

HINT. Show that the relations for r and s follow from the relations for a and b and conversely, the relations for a and b follow from those for r and s .

Solution. If $a^2 = b^2 = (ab)^n = 1$, put $r = ab$ and $s = a$. Then

$$r^n = (ab)^n = 1, \quad s^2 = a^2 = 1, \quad rs = (ab)a = a(ba) = a(b^{-1}a^{-1}) = a(ab)^{-1} = sr^{-1}.$$

Conversely, put $a = s$ and $b = sr$. The calculations are similar. ■

Exercise 1.2.8

Find the order of the **cyclic subgroup** of D_{2n} generated by r .

Solution. Since $|r| = n$, the cyclic subgroup generated by r has order n by [Exercise 1.1.35](#). ■

Exercise 1.2.9

Let G be the group of rigid motions in $\mathbb{R}^3$ of a tetrahedron. Show that $|G| = 12$.

Solution. Label the vertices of a tetrahedron 1 through 4. Vertex 1 has 4 choices, vertex 2 has 3 remaining choices, and vertices 3 and 4 are determined after. It follows that $4(3) = 12$ symmetries. ■

Exercise 1.2.10

Let G be the group of rigid motions in $\mathbb{R}^3$ of a cube. Show that $|G| = 24$.

Solution. 8 choices for vertex 1 and 3 adjacent vertices for vertex 2; $8(3) = 24$ symmetries. ■

Exercise 1.2.11

Let G be the group of rigid motions in $\mathbb{R}^3$ of an octahedron. Show that $|G| = 24$.

Solution. 6 choices for vertex 1 and 4 adjacent vertices for vertex 2; $6(4) = 24$ symmetries. ■

Exercise 1.2.12

Let G be the group of rigid motions in $\mathbb{R}^3$ of a dodecahedron. Show that $|G| = 60$.

Solution. 20 choices for vertex 1 and 3 adjacent vertices for vertex 2; $20(3) = 60$ symmetries. ■

Exercise 1.2.13

Let G be the group of rigid motions in $\mathbb{R}^3$ of an icosahedron. Show that $|G| = 60$.

Solution. 12 choices for vertex 1 and 5 adjacent vertices for vertex 2; $12(5) = 60$ symmetries. ■

Exercise 1.2.14

Find a set of generators for $\mathbb{Z}$.

Solution. Any integer is of the form $n1$, so $\mathbb{Z} = \langle 1 \rangle$. ■

Exercise 1.2.15

Find a set of generators and relations for $\mathbb{Z}/n\mathbb{Z}$.

Solution. Each element of $\mathbb{Z}/n\mathbb{Z}$ is of the form $k\bar{1}$ for $0 \leq k < n$. Then $\mathbb{Z}/n\mathbb{Z} = \langle \bar{1} \mid n\bar{1} = \bar{0} \rangle$. ■

Exercise 1.2.16

Show that the group $\langle x_1, y_1 \mid x_1^2 = y_1^2 = (x_1 y_1)^2 = 1 \rangle$ is the dihedral group D_4 (where x_1 may be replaced by the letter r and y_1 by s).

HINT. Show that the last relation is the same as $x_1 y_1 = y_1 x_1^{-1}$.

Solution. Note that $y_1^2 = 1$, then $y_1^{-1} = y_1$. Then

$$(x_1 y_1)^2 = 1 \iff x_1 y_1 x_1 y_1 = 1 \iff x_1 y_1 x_1 = y_1^{-1} = y_1 \iff x_1 y_1 = y_1 x_1^{-1}$$

Then the relations of D_4 are satisfied by x_1 and y_1 if we set $r = x_1$ and $s = y_1$. ■

Exercise 1.2.17

Let X_{2n} be the group whose presentation is displayed in (1.2).

$$X_{2n} = \langle x, y \mid x^n = y^2 = 1, xy = yx^2 \rangle$$

- (a) Show that if $n = 3k$, then X_{2n} has order 6, and it has the same generators and relations as D_6 when x is replaced by r and y by s .
- (b) Show that if $(3, n) = 1$, then x satisfies the additional relation: $x = 1$. In this case deduce that X_{2n} has order 2.

HINT. Use the facts that $x^n = 1$ and $x^3 = 1$.

Solution.

- (a) If $n = 3k$, then

$$X_{2n} = X_{6k} = \langle x, y \mid x^{3k} = y^2 = 1, xy = yx^2 \rangle$$

As shown in the text, $x^3 = 1$. Assume the relations of X_{6k} hold. Set $r = x$ and $s = y$, so that $r^3 = x^3 = 1$, $s^2 = y^2 = 1$, and

$$rs = xy = yx^2 = sr^2 = sr^{-1}.$$

Now suppose the relations of D_6 hold. Then

$$x^n = r^{3k} = (r^3)^k = 1^k = 1, \quad y^2 = s^2 = 1$$

and

$$xy = rs = sr^{-1} = sr^2 = yx^2$$

Thus, the two presentations are equivalent, and $|X_{6k}| = |D_6| = 6$.

- (b) Since $(3, n) = 1$, there exist integers a and b such that $3a + nb = 1$. From the relations of X_{2n} , we have $x^3 = 1$ and $x^n = 1$. Then

$$x = x^{3a+nb} = (x^3)^a (x^n)^b = 1^a 1^b = 1$$

so that $X_{2n} = 1, y$ where $y^2 = 1$. It follows that $|X_{2n}| = 2$. ■

Exercise 1.2.18

Let Y be the group whose presentation is displayed in (1.3).

$$Y = \langle u, v \mid u^4 = v^3 = 1, uv = v^2u^2 \rangle.$$

- (a) Show that $v^2 = v^{-1}$.

HINT. Use the relation $v^3 = 1$.

- (b) Show that v commutes with u^3 .

HINT. Show that $v^2u^3v = u^3$ by writing the left-hand side as $(v^2u^2)(uv)$ and using the relations to reduce this to the right-hand side. Then use part (a).

- (c) Show that v commutes with u .

HINT. Show that $u^9 = u$ and then use part (b).

- (d) Show that $uv = 1$.

HINT. Use part (c) and the last relation.

- (e) Show that $u = 1$. Deduce that $v = 1$, and conclude that $Y = 1$.

HINT. Use part (d) and the equation $u^4v^3 = 1$.

Solution.

(a) $v^3 = 1 \implies v^2 = v^{-1}$.

(b) From $uv = v^2u^2$, we have $vuv = u^2$ so that $vu = u^2v^{-1} = u^2v^2$. It follows that

$$v^2u^3v = (v^2u^2)(uv) = (uv)(uv) = u(vu)v = u(u^2v^2)v = u^3.$$

(c) It follows that $u^9 = (u^4)^2u = u$. Then $uv = (u^3)^3v = v(u^3)^3 = vu$.

(d) $uv = v^2u^2 = u^2v^2$. Then $1 = uv$.

(e) $u^4v^3 = u(uv)^3 = u1^3 = 1$. Then $1v = 1$, and $u = v = 1$, so $Y = 1$. ■

1.3 Symmetric Groups

Exercise 1.3.1

Let σ be the permutation

$$1 \mapsto 3 \quad 2 \mapsto 4 \quad 3 \mapsto 5 \quad 4 \mapsto 2 \quad 5 \mapsto 1$$

and let τ be the permutation

$$1 \mapsto 5 \quad 2 \mapsto 3 \quad 3 \mapsto 2 \quad 4 \mapsto 4 \quad 5 \mapsto 1$$

Find the cycle decompositions of each of the following permutations: $\sigma, \tau, \sigma^2, \sigma\tau, \tau\sigma$, and $\tau^2\sigma$.

Solution. We have $\sigma = (1\ 3\ 5)(2\ 4)$ and $\tau = (1\ 5)(2\ 3)$. The rest is straightforward. ■

Exercise 1.3.2

Let σ be the permutation

$$\begin{array}{ccccc} 1 \mapsto 13 & 2 \mapsto 2 & 3 \mapsto 15 & 4 \mapsto 14 & 5 \mapsto 10 \\ 6 \mapsto 6 & 7 \mapsto 12 & 8 \mapsto 3 & 9 \mapsto 4 & 10 \mapsto 1 \\ 11 \mapsto 7 & 12 \mapsto 9 & 13 \mapsto 5 & 14 \mapsto 11 & 15 \mapsto 8 \end{array}$$

and let τ be the permutation

$$\begin{array}{ccccc} 1 \mapsto 14 & 2 \mapsto 9 & 3 \mapsto 10 & 4 \mapsto 2 & 5 \mapsto 12 \\ 6 \mapsto 6 & 7 \mapsto 5 & 8 \mapsto 11 & 9 \mapsto 15 & 10 \mapsto 3 \\ 11 \mapsto 8 & 12 \mapsto 7 & 13 \mapsto 4 & 14 \mapsto 1 & 15 \mapsto 13 \end{array}$$

Find the cycle decompositions of the following permutations: $\sigma, \tau, \sigma^2, \sigma\tau, \tau\sigma$, and $\tau^2\sigma$.

Solution. $\sigma = (1\ 13\ 5\ 10)(3\ 15\ 8)(4\ 14\ 11\ 7\ 12\ 9)$ and $\tau = (1\ 14)(2\ 9\ 15\ 13\ 4)(3\ 10)(5\ 12\ 7)(8\ 11)$. Again, the rest is straightforward. ■

Exercise 1.3.3

For each of the permutations whose cycle decompositions were computed in the preceding two exercises, compute its order.

Solution.

$$\begin{array}{ll} |\sigma| = 6, 12 & |\tau| = 2, 30 \\ |\sigma^2| = 3, 6 & |\sigma\tau| = 4, 6 \\ |\tau\sigma| = 4, 6 & |\tau^2\sigma| = 6, 13 \end{array}$$

Exercise 1.3.4

Compute the order of each of the elements in the following groups:

(a) S_3 (b) S_4

Solution.

(a)

Permutation	Order
1	1
(12)	2
(13)	2
(23)	2
(123)	3
(132)	3

(b)

Permutation	Order	Permutation	Order	Permutation	Order	Permutation	Order
1	1	(34)	2	(143)	3	(1342)	4
(12)	2	(123)	3	(234)	3	(1423)	4
(13)	2	(124)	3	(243)	3	(1432)	4
(14)	2	(132)	3	(1234)	4	(12)(34)	2
(23)	2	(134)	3	(1243)	4	(13)(24)	2
(24)	2	(142)	3	(1324)	4	(14)(23)	2

■

Exercise 1.3.5Find the order of $(1\ 12\ 8\ 10\ 4)(2\ 13)(5\ 11\ 7)(6\ 9)$.**Solution.** There are cycles of lengths 5, 2, 3, and 2. Then the order is $\text{lcm}(5, 2, 3, 2) = 30$. ■**Exercise 1.3.6**Write out the cycle decomposition of each element of order 4 in S_4 .**Solution.** See Exercise 1.3.4. ■**Exercise 1.3.7**Write out the cycle decomposition of each element of order 2 in S_4 .**Solution.** See Exercise 1.3.4. ■**Exercise 1.3.8**Prove that if Ω is a finite set, then S_Ω is a finite group. Prove that if $\Omega = \{1, 2, 3, \dots\}$, then S_Ω is an infinite group (do not say $\infty! = \infty$).**Solution.** For each $n \in \mathbb{Z}^+$, consider the permutation $\sigma_n = (1\ n)$. Then $\sigma_n \in S_\Omega$ and $\sigma_n \neq \sigma_m$ for $n \neq m$, since they move different elements. Hence, S_Ω contains an infinite collection of distinct elements, so S_Ω is infinite. ■

Exercise 1.3.9

Let σ be an m -cycle. For which positive integers i is σ^i also an m -cycle?

- (a) Let σ be the 12-cycle $(1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 10\ 11\ 12)$. For which positive integers i is σ^i also a 12-cycle?
- (b) Let τ be the 8-cycle $(1\ 2\ 3\ 4\ 5\ 6\ 7\ 8)$. For which positive integers i is τ^i also an 8-cycle?
- (c) Let ω be the 14-cycle $(1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 10\ 11\ 12\ 13\ 14)$. For which positive integers i is ω^i also a 14-cycle?

AUTHOR'S NOTE. The reader should notice a pattern: integers i such that $(m, i) = 1$ produce an m -cycle, while integers i such that $(m, i) = k$ produce k disjoint m/k -cycles.

Solution.

- (a) Computing the cycle for each integer 0 to 12:

$$\begin{array}{ll} \sigma^1 = \sigma & \sigma^7 = (1\ 8\ 3\ 10\ 5\ 12\ 7\ 2\ 9\ 4\ 11\ 6) \\ \sigma^2 = (1\ 3\ 5\ 7\ 9\ 11)(2\ 4\ 6\ 8\ 10\ 12) & \sigma^8 = (1\ 9\ 5)(2\ 10\ 6)(3\ 11\ 7)(4\ 12\ 8) \\ \sigma^3 = (1\ 4\ 7\ 10)(2\ 5\ 8\ 11)(3\ 6\ 9\ 12) & \sigma^9 = (1\ 10\ 7\ 4)(2\ 11\ 8\ 5)(3\ 12\ 9\ 6) \\ \sigma^4 = (1\ 5\ 9)(2\ 6\ 10)(3\ 7\ 11)(4\ 8\ 12) & \sigma^{10} = (1\ 11\ 9\ 7\ 5\ 3)(2\ 12\ 10\ 8\ 6\ 4) \\ \sigma^5 = (1\ 6\ 11\ 4\ 9\ 2\ 7\ 12\ 5\ 10\ 3\ 8) & \sigma^{11} = (1\ 12\ 11\ 10\ 9\ 8\ 7\ 6\ 5\ 4\ 3\ 2) \\ \sigma^6 = (1\ 7)(2\ 8)(3\ 9)(4\ 10)(5\ 11)(6\ 12) & \sigma^{12} = \text{id} \end{array}$$

Hence, the set of integers that produce a 12-cycle is the set $\{i + 12k \mid i \in \{1, 5, 7, 11\}, k \in \mathbb{Z}\}$.

- (b) $\{i + 8k \mid i \in \{1, 3, 5, 7\}, k \in \mathbb{Z}\}$.

- (c) $\{i + 14k \mid i \in \{1, 3, 5, 9, 11, 13\}, k \in \mathbb{Z}\}$. ■

Exercise 1.3.10

Prove that if σ is the m -cycle $(a_1\ a_2\ \dots\ a_m)$, then for all $i \in \{1, 2, \dots, m\}$, $\sigma^i(a_k) = a_{k+i}$, where $k + i$ is replaced by its least residue mod m when $k + i > m$. Deduce that $|\sigma| = m$.

Solution. Assume all subscripts of a are taken mod m . Inducting on $i = 1$, the base case is clear. If $\sigma^i(a_k) = a_{k+i}$ for $1 \leq i \leq m - 1$, then

$$\sigma^{i+1}(a_k) = \sigma(\sigma^i(a_k)) = \sigma(a_{k+i}) = a_{k+i+1}.$$

The result follows by induction.

To see that $|\sigma| = m$, note that $\sigma^m(a_k) = a_{k+m} = a_k$ for all $k \in \{1, 2, \dots, m\}$, so that $\sigma^m = \text{id}$. If $|\sigma| = i < m$, then $\sigma^i(a_1) = a_{1+i} = a_1$, which implies that $i \equiv 0 \pmod{m}$. Hence, $|\sigma| = m$. ■

Exercise 1.3.11

Let σ be the m -cycle $(1\ 2\ \dots\ m)$. Show that σ^i is also an m -cycle if and only if i is relatively prime to m .

Solution.

($\Rightarrow$) Suppose $(m, i) = k > 1$. If $i > m$, replace i with $i \bmod m$ since $\sigma^i = \sigma^{i \bmod m}$. Then $i = k\ell$ for $\ell \in \mathbb{Z}^+$. It is clear that $(\sigma^i)^{m/k} = \sigma^{\ell m} = \text{id}$, so that $|\sigma^i| \leq m/k < m$. Hence, σ^i is not an m -cycle.

($\Leftarrow$) Suppose that $(m, i) = 1$. Denote $x_m \equiv x \pmod{m}$. We claim that the integers

$$1_m, (1+i)_m, (1+2i)_m, \dots, (1+(m-1)i)_m$$

are all distinct. Suppose $(1+xi)_m = (1+yi)_m$ for some $0 \leq x, y \leq m-1$ with $x \neq y$. Then $i(x-y) \equiv 0 \pmod{m}$. Since $(m, i) = 1$, then $m \mid (x-y)$. However, since $1-m \leq x-y \leq m-1$, then the only choice for $x-y$ is 0, or that $x = y$. Then we have exactly m distinct integers so that

$$\sigma^i = (1 \ (1+i)_m \ (1+2i)_m \ \dots \ (1+(m-1)i)_m).$$

Thus, σ^i is an m -cycle. ■

Exercise 1.3.12

- (a) If $\tau = (1\ 2)(3\ 4)(5\ 6)(7\ 8)(9\ 10)$, determine whether there is an n -cycle ($n \geq 10$) with $\tau = \sigma^k$ for some integer k .
 (b) If $\tau = (1\ 2)(3\ 4\ 5)$, determine whether there is an n -cycle ($n \geq 5$) with $\tau = \sigma^k$ for some integer k .

Solution.

- (a) We first observe that for any n -cycle σ , then σ^k sends any element a_i to a_{i+k} , where $i+k$ is replaced by its least residue mod n when $i+k > n$. In this specific problem, τ consists of 2-cycles, which means that σ^k must send each element to itself after two applications. Then $2k \equiv 0 \pmod{n}$, but $k \not\equiv 0 \pmod{n}$ since $\sigma^k \neq 1$. It follows that $n \mid 2k$ but $n \nmid k$, so it must be that $n = 2k$. We see that σ^k will be a product of k 2-cycles. However, τ is a product of 5 2-cycles, so $k = 5$ and $n = 10$. Thus, there exists an n -cycle σ such that $\sigma^5 = \tau$ when $n = 10$, and $\sigma = (1\ 6\ 2\ 7\ 3\ 8\ 4\ 9\ 5\ 10)$.
 (b) We first prove that $(ac, bc) = c(a, b)$ for $a, b, c \in \mathbb{Z}$.

Lemma: L

Proof. Let $d = (a, b)$ and $d' = (ac, bc)$. Then there exist $s, t \in \mathbb{Z}$ such that $d = as + bt$. Consider $cd = acs + bct$. Then $d' = (ac, bc)$ implies that $d' \mid cd$. Since $d \mid a$ and $d \mid b$, then $cd \mid ac$ and $cd \mid bc$, so that $cd \mid (ac, bc) = d'$. Then $d' = cd$ or $d' = c(a, b)$. ■

Assume for contradiction that there exists an n -cycle σ and k such that $\sigma^k = \tau$. Then $\tau^2(1) = 1$, hence $\sigma^{2k}(1) = 1$, or $2k \equiv 0 \pmod{n}$. Moreover, $\tau^3(3) = 3$, hence $\sigma^{3k}(3) = 3$, or $3k \equiv 0 \pmod{n}$. Then $n \mid 2k$ and $n \mid 3k$, so that $n \mid (2k, 3k)$. By the lemma, $(2k, 3k) = k(2, 3) = k$, so that $n \mid k$. However, $\sigma^k \neq 1$, so $n \nmid k$. This is a contradiction, so there does not exist an n -cycle σ such that $\sigma^k = \tau$. ■

Exercise 1.3.13

Show that an element has order 2 in S_n if and only if its cycle decomposition is a product of commuting 2-cycles.

Solution.

- ($\Rightarrow$) Let $\sigma \in S_n$ with $|\sigma| = 2$. Let $(a_1\ a_2\ \dots\ a_m)$ be one of the cycles in its cycle decomposition. Since $\sigma \neq \text{id}$ and $\sigma^2 = 1$, then $\sigma(a_1) = a_2$ and $\sigma(a_2) = a_1$. Then $m = 2$, and the cycle is a 2-cycle.
 ($\Leftarrow$) Suppose σ 's cycle decomposition is a product of commuting 2-cycles $(a_1\ b_1)(a_2\ b_2)\dots(a_k\ b_k)$. Since each 2-cycle has order 2, and the 2-cycles commute, then $\sigma^2 = \text{id}$. Because $\sigma \neq \text{id}$, then $|\sigma| = 2$. ■

Exercise 1.3.14

Let p be a prime. Show that an element has order p in S_n if and only if its cycle decomposition is a product of commuting p -cycles. Show by an explicit example that this need not be the case if p is not prime.

Solution.

($\Rightarrow$) Let $\sigma \in S_n$ with $|\sigma| = p$. Let $(a_1 a_2 \dots a_m)$ be one of the cycles in its cycle decomposition. By [Exercise 1.3.10](#), we know $\sigma^i(a_i) = a_{1+i}$ where $1+i$ is taken mod m when $1+i > m$. Then $\sigma^p(a_1) = a_{1+p} = a_1$, so that $m \mid p$. Since p is prime, then $m = p$, hence any cycle in the cycle decomposition of σ is a p -cycle.

($\Leftarrow$) Let $\sigma = \tau_1 \tau_2 \dots \tau_k$ be the cycle decomposition of σ , where each τ_i is a p -cycle. Since the cycles are disjoint, they commute. Moreover, τ_i^t is a non-trivial p -cycle for $t < p$ by [Exercise 1.3.11](#). Then $\sigma^t = \tau_1^t \tau_2^t \dots \tau_k^t$ is a product of non-trivial p -cycles for $t < p$, so that $\sigma^t \neq 1$. However, $\sigma^p = \tau_1^p \tau_2^p \dots \tau_k^p = 1$, so that $|\sigma| = p$.

For a counterexample when p is not prime, consider $\sigma = (1\ 2\ 3)(4\ 5)$ in S_5 . Then $|\sigma| = 6$, but the cycle decomposition consists of a 3-cycle and a 2-cycle. ■

Exercise 1.3.15

Prove that the order of an element in S_n equals the least common multiple of the lengths of the cycles in its cycle decomposition.

Solution. Let $\sigma \in S_n$ have cycle decomposition $\tau_1 \tau_2 \dots \tau_k$, where each τ_i is a cycle of length m_i . Then $|\tau_i| = m_i$ by [Exercise 1.3.10](#). Since the cycles are disjoint, they commute, so that $\sigma^t = \tau_1^t \tau_2^t \dots \tau_k^t$. Then $\sigma^t = 1$ if and only if $\tau_i^t = 1$ for all i , which occurs if and only if $m_i \mid t$ for all i . Hence, the order of σ is the least common multiple of the lengths of the cycles in its cycle decomposition. ■

Exercise 1.3.16

Show that if $n \geq m$ then the number of m -cycles in S_n is given by

$$\frac{n(n-1)(n-2) \dots (n-m+1)}{m}.$$

HINT. Count the number of ways of forming an m -cycle and divide by the number of representations of a particular m -cycle.

Solution. Note that in an m -cycle, there are m integers to write. The first integer has n choices, the second $n-1$ choices. Continuing on, then the m -th integer has $n-m+1$ choices. However, each choice of m integers has m equivalent representations (such as $(1\ 2\ 3\ 4)$ being equivalent to $(2\ 3\ 4\ 1)$ by shifting each integer to the left one place). Then take the product $n(n-1)(n-2) \dots (n-m+1)$ and divide by m . ■

Exercise 1.3.17

Show that if $n \geq 4$ then the number of permutations in S_n which are the product of two disjoint 2-cycles is

$$\frac{n(n-1)(n-2)(n-3)}{8}.$$

Solution. The first 2-cycle has $n(n-1)/2$ choices, and the second 2-cycle has $(n-2)(n-3)/2$ choices. Moreover, there are 2 representations for the same set of 2-cycles, so multiply the given quantities and divide by 2. ■

Exercise 1.3.18

Prove that if $n \geq 2m$ then the number of permutations in S_n which are the product of m disjoint 2-cycles is given by Find all numbers n such that S_5 contains an element of order n .

Solution. S_5 contains elements of order 1 through 5. Moreover, we can have a 2 and 3-cycle together whose LCM is 6. Then $n = 1, 2, 3, 4, 5, 6$. ■

Exercise 1.3.19

Find all numbers n such that S_7 contains a number an element of order n .

Solution. We have n from 1 to 7. We can have 2 and 5-cycles and 3 and 4-cycles. Then $n = 10$ and 12. ■

Exercise 1.3.20

Find a set of generators and relations for S_3 .

Solution. Recall that $S_3 = \{1, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$. Moreover, no element in S_3 has order 6 so that no element generates S_3 . It must be that there are at least 2 generators. Put $\alpha = (1\ 2)$ and $\beta = (1\ 3)$. Then $\alpha^2 = \beta^2 = 1$. Then $|\alpha| = |\beta| = 2$, and $\alpha\beta = (1\ 3\ 2)$, $\beta\alpha = (1\ 2\ 3)$, and $\alpha\beta\alpha = (2\ 3)$. Since $\alpha^2 = \beta^2 = 1$ is not enough to determine the order of $\alpha\beta$ (because $\alpha\beta \neq \beta\alpha$), then we must add $(\alpha\beta)^2 = 1$. Then we have the presentation $S_3 = \langle \alpha, \beta \mid \alpha^2 = \beta^2 = (\alpha\beta)^3 = 1 \rangle$. ■

1.4 Matrix Groups

Let F be a field and let $n \in \mathbb{Z}^+$.

Exercise 1.4.1

Prove that $|\mathrm{GL}_2(\mathbb{F}_2)| = 6$.

Solution. Direct calculation gives:

$$\mathrm{GL}_2(\mathbb{F}_2) = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \right\}. \quad \blacksquare$$

Exercise 1.4.2

Write out all the elements of $\mathrm{GL}_2(\mathbb{F}_2)$ and compute the order of each element.

Solution. By direct calculation, we obtain the following:

$$\begin{array}{c|c|c|c|c|c|c} A & \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} & \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \\ \hline |A| & 1 & 2 & 2 & 2 & 3 & 3 \end{array} \quad \blacksquare$$

Exercise 1.4.3

Show that $\mathrm{GL}_2(\mathbb{F}_2)$ is non-Abelian.

Solution. It is trivial to see that the following two matrices do not commute, hence $\mathrm{GL}_2(\mathbb{F}_2)$ is non-Abelian.

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}. \quad \blacksquare$$

Exercise 1.4.4

Show that if n is not prime then $\mathbb{Z}/n\mathbb{Z}$ is not a field.

Solution. Since n is not prime, it must be composite. Then there exist $a, b \in \mathbb{Z}^+$ such that $1 < a, b < n$ and $n = ab$. Moreover, $(a, n) \neq 1$ since $a \mid n$. Then a has no multiplicative inverse in $\mathbb{Z}/n\mathbb{Z}$, so $\mathbb{Z}/n\mathbb{Z}$ is not a field. $\blacksquare$

Exercise 1.4.5

Show that $\mathrm{GL}_n(F)$ is a finite group if and only if F has a finite number of elements.

Solution.

($\Rightarrow$) If $\mathrm{GL}_n(F)$ is finite, the subset $\{\alpha I \mid \alpha \in F\}$ must also be finite, which implies that F has a finite number of elements.

($\Leftarrow$) Suppose $|F| = q$. Then an $n \times n$ matrix over F has q^{n^2} possible entries. Since $\mathrm{GL}_n(F)$ is a subset of the set of all $n \times n$ matrices over F , it follows that $\mathrm{GL}_n(F)$ is finite. $\blacksquare$

Exercise 1.4.6

If $|F| = q$ is finite, prove that $|\mathrm{GL}_n(F)| < q^{n^2}$.

Solution. See $(\Leftarrow)$ in [Exercise 1.4.5](#). ■

Exercise 1.4.7

Let p be a prime. Prove that the order of $\mathrm{GL}_2(\mathbb{F}_p)$ is $p^4 - p^3 - p^2 + p$.

REMARK. Do not just quote the order formula in this section.

HINT. Subtract the number of 2×2 matrices which are not invertible from the total number of 2×2 matrices over $\mathbb{F}_p$. You may use the fact that a 2×2 matrix is not invertible if and only if one row is a multiple of the other.

Solution. Observe that there are p^4 total 2×2 matrices over $\mathbb{F}_p$. The number of non-invertible matrices is as follows:

- If the first row is $(0, 0)$, then there are p^2 choices for the second row.
- If the first row is $(a, b) \neq (0, 0)$, then there are p choices for the second row, since it must be a multiple of the first row. There are $p^2 - 1$ choices for the first row.

Then the number of non-invertible matrices is $p^2 + p(p^2 - 1) = p^3 + p^2 - p$. Subtracting from the total number of matrices, we have the desired number. ■

Exercise 1.4.8

Show that $\mathrm{GL}_n(F)$ is non-Abelian for any $n \geq 2$ and any F .

Solution. Consider the subset $S = \{0, 1\}$ of F . Then $\mathrm{GL}_n(S)$ is a subset of $\mathrm{GL}_n(F)$. We proceed by induction on this subset. For $n = 2$, we have $\mathrm{GL}_2(S)$ is non-Abelian by [Exercise 1.4.3](#). Suppose that $\mathrm{GL}_n(S)$ is non-Abelian for some $n \geq 2$. Then there exist $A, B \in \mathrm{GL}_n(S)$ such that $AB \neq BA$. Consider the matrices

$$A_0 = \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}, B_0 = \begin{pmatrix} B & 0 \\ 0 & 1 \end{pmatrix} \in \mathrm{GL}_{n+1}(S).$$

Observe that $A_0 B_0$ contains AB as a block matrix, while $B_0 A_0$ contains BA as a block matrix. Since $AB \neq BA$, then $A_0 B_0 \neq B_0 A_0$. Hence, $\mathrm{GL}_{n+1}(S)$ is non-Abelian. By induction, $\mathrm{GL}_n(S)$ is non-Abelian for all $n \geq 2$. Since $\mathrm{GL}_n(S) \subseteq \mathrm{GL}_n(F)$, then $\mathrm{GL}_n(F)$ is non-Abelian for all $n \geq 2$. ■

Exercise 1.4.9

Prove that the binary operation of matrix multiplication of 2×2 matrices with real number entries is associative.

Solution. The direct proof is laborious. Instead, associate with any 2×2 matrix some linear map. Since composition of linear maps is associative, the result follows. ■

Exercise 1.4.10

Let

$$G = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \mid a, b, c \in \mathbb{R}, a \neq 0, c \neq 0 \right\}$$

(a) Compute the product of

$$\begin{pmatrix} a_1 & b_1 \\ 0 & c_1 \end{pmatrix} \text{ and } \begin{pmatrix} a_2 & b_2 \\ 0 & c_2 \end{pmatrix}$$

to show that G is closed under matrix multiplication.

(b) Find the matrix inverse of

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$$

and deduce that G is closed under inverses.(c) Deduce that G is a **subgroup** of $\text{GL}_2(\mathbb{R})$.(d) Prove that the set of elements of G whose two diagonal entries are equal (i.e., $a = c$) is also a **subgroup** of $\text{GL}_2(\mathbb{R})$.**Solution.**

(a)

$$\begin{pmatrix} a_1 & b_1 \\ 0 & c_1 \end{pmatrix} \begin{pmatrix} a_2 & b_2 \\ 0 & c_2 \end{pmatrix} = \begin{pmatrix} a_1 a_2 & a_1 b_2 + b_1 c_2 \\ 0 & c_1 c_2 \end{pmatrix}$$

Since $a_i \neq 0$ and $c_i \neq 0$, then $a_1 a_2 \neq 0$ and $c_1 c_2 \neq 0$ so that G is closed under matrix multiplication.(b) The determinant is ac . We then have the inverse

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}^{-1} = \frac{1}{ac} \begin{pmatrix} c & -b \\ 0 & a \end{pmatrix} = \begin{pmatrix} 1/a & -b/ac \\ 0 & 1/c \end{pmatrix}$$

(c) Since G is closed under inverses and matrix multiplication, the result follows.(d) Let $H = \{A \in G \mid a = c\}$. Then

$$\begin{pmatrix} a_1 & b_1 \\ 0 & a_1 \end{pmatrix} \begin{pmatrix} a_2 & b_2 \\ 0 & a_2 \end{pmatrix} = \begin{pmatrix} a_1 a_2 & a_1 b_2 + b_1 a_2 \\ 0 & a_1 a_2 \end{pmatrix} \in H$$

and

$$\begin{pmatrix} a & b \\ 0 & a \end{pmatrix}^{-1} = \frac{1}{a^2} \begin{pmatrix} a & -b \\ 0 & a \end{pmatrix} = \begin{pmatrix} 1/a & -b/a^2 \\ 0 & 1/a \end{pmatrix} \in H$$

Then H is closed under inverses and matrix multiplication, hence it is a subgroup of $\text{GL}_2(\mathbb{R})$. ■

Exercise 1.4.11

Let

$$H(F) = \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \mid a, b, c \in F \right\},$$

called the **Heisenberg group** over F . Let

$$X = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \text{ and } Y = \begin{pmatrix} 1 & d & e \\ 0 & 1 & f \\ 0 & 0 & 1 \end{pmatrix}$$

be elements of $H(F)$.

- Compute the matrix product XY and deduce that $H(F)$ is closed under matrix multiplication. Exhibit explicit matrices such that $XY \neq YX$ (so that $H(F)$ is always non-Abelian).
- Find an explicit formula for the matrix inverse X^{-1} and deduce that $H(F)$ is closed under inverses.
- Prove the associative law for $H(F)$ and deduce that $H(F)$ is a group of order $|F|^3$. (Do not assume that matrix multiplication is associative.)
- Find the order of each element of the finite group $H(\mathbb{Z}/2\mathbb{Z})$.
- Prove that every non-identity of the group $H(\mathbb{R})$ has infinite order.

AUTHOR'S NOTE. The choice of rewriting the matrix as a tuple is simply a time saver/ease of notation for the sake of computation. However, it does allow one to encode $H(F)$ in the multiplication law of the field.

Solution.

- Calculating the product, we have

$$XY = \begin{pmatrix} 1 & d+a & e+af+b \\ 0 & 1 & f+c \\ 0 & 0 & 1 \end{pmatrix} \in H(F)$$

It is trivial to exhibit non-commutative matrices, hence $H(F)$ is non-Abelian.

- From part (a), we know G is closed under matrix multiplication. For ease of notation, represent a matrix as a tuple: $X = (a, b, c)$ and $Y = (d, e, f)$. Then we may represent matrix multiplication in $H(F)$ as

$$(a, b, c)(d, e, f) = (a + d, af + b + e, c + f)$$

Now we want to find (x, y, z) such that $(a, b, c)(x, y, z) = (0, 0, 0)$. Then $(a + x, af + b + y, c + z) = (0, 0, 0)$ implies $x = -a$, $y = -b + ac$, and $z = -c$. It follows that $X^{-1} = (-a, -b + ac, -c)$.

- For some $(a, b, c) \in H(F)$, we know that there are $|F|$ choices for each of a, b, c . Then $|H(F)| = |F|^3$. Now let $(a, b, c), (d, e, f), (g, h, i) \in H(F)$. Then we have

$$\begin{aligned} ((a, b, c)(d, e, f))(g, h, i) &= (a + d, af + b + e, c + f)(g, h, i) \\ &= (a + d + g, (a + d)i + af + b + e + h, c + f + i) \\ &= (a + d + g, ai + di + af + b + e + h, c + f + i) \\ &= (a, b, c)(d + g, e + h, f + i) \\ &= (a, b, c)((d, e, f)(g, h, i)) \end{aligned}$$

(d) The elements of $H(\mathbb{Z}/2\mathbb{Z})$ are

$$(0, 0, 0), (1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 0), (1, 0, 1), (0, 1, 1), (1, 1, 1).$$

Since $(a, b, c)^2 = (2a, 2b + ac, 2c) = (0, ac, 0)$ in $\mathbb{Z}/2\mathbb{Z}$, then we have two cases:

- If $ac = 0$, then at least one of a or c is 0, while b varies. These are the matrices that have order 2:

$$(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 0), (0, 1, 1).$$

- If $ac = 1$, then both a and c are 1, while b varies. These are the matrices that have order 4:

$$(1, 0, 1), (1, 1, 1).$$

(e) Let $X \in H(\mathbb{R})$ be non-identity. It is easy to see by induction that

$$X^n = (na, n(n-1)ac/2 + nb, nc).$$

Since $X \neq I_3$, then at least one of a, b, c is nonzero. It is clear that $a \neq 0$ or $c \neq 0$ implies $X^n \neq I_3$. Now suppose $a = c = 0$ and $b \neq 0$. Then $X^n = (0, nb, 0)$. Since $b \neq 0$, then $nb \neq 0$ for all $n \in \mathbb{Z}^+$. Hence, $X^n \neq I_3$ for all $n \in \mathbb{Z}^+$. Therefore, every non-identity element of $H(\mathbb{R})$ has infinite order. ■

1.5 The Quaternion Group

Exercise 1.5.1

Compute the order of each of the elements in Q_8 .

Solution.

a	1	-1	i	$-i$	j	$-j$	k	$-k$
$ a $	1	2	4	4	4	4	4	4

■

Exercise 1.5.2

Write out the group tables for S_3 , D_8 , and Q_8 .

AUTHOR'S NOTE. *Tedious, but straightforward.*

Solution.

S_3 :

$\circ$	1	(1 2)	(1 3)	(2 3)	(1 2 3)	(1 3 2)
1	1	(1 2)	(1 3)	(2 3)	(1 2 3)	(1 3 2)
(1 2)	(1 2)	1	(1 2 3)	(1 3 2)	(1 3)	(2 3)
(1 3)	(1 3)	(1 3 2)	1	(1 2 3)	(2 3)	(1 2)
(2 3)	(2 3)	(1 2 3)	(1 3 2)	1	(1 2)	(1 3)
(1 2 3)	(1 2 3)	(2 3)	(1 2)	(1 3)	(1 3 2)	1
(1 3 2)	(1 3 2)	(1 3)	(2 3)	(1 2)	1	(1 2 3)

D_8 :

$\circ$	1	r	r^2	r^3	s	sr	sr^2	sr^3
1	1	r	r^2	r^3	s	sr	sr^2	sr^3
r	r	r^2	r^3	1	sr^3	s	sr	sr^2
r^2	r^2	r^3	1	r	sr^2	sr^3	s	sr
r^3	r^3	1	r	r^2	sr	sr^2	sr^3	s
s	s	sr	sr^2	sr^3	1	r	r^2	r^3
sr	sr	sr^2	sr^3	s	r^3	1	r	r^2
sr^2	sr^2	sr^3	s	sr	r^2	r^3	1	r
sr^3	sr^3	s	sr	sr^2	r	r^2	r^3	1

■

Q_8 :

$\cdot$	1	-1	i	$-i$	j	$-j$	k	$-k$
1	1	-1	i	$-i$	j	$-j$	k	$-k$
-1	-1	1	$-i$	i	$-j$	j	$-k$	k
i	i	$-i$	1	1	k	$-k$	j	$-j$
$-i$	$-i$	i	1	-1	$-k$	k	$-j$	j
j	j	$-j$	$-k$	k	-1	1	i	$-i$
$-j$	$-j$	j	k	$-k$	1	-1	$-i$	i
k	k	$-k$	j	$-j$	$-i$	i	-1	1
$-k$	$-k$	k	$-j$	j	i	$-i$	1	-1

Exercise 1.5.3

Find a set of generators and relations for Q_8 .

AUTHOR'S NOTE. *One may notice I dropped k . Presentations should utilize minimal generators for efficiency, and since k can be expressed in terms of i and j , it is not necessary to include it.*

Solution. $\langle i, j \mid i^4 = 1, i^2 = j^2, ij = -ji \rangle$. ■

1.6 Homomorphisms and Isomorphisms

Exercise 1.6.1

Let $\varphi : G \rightarrow H$ be a homomorphism.

- (a) Prove that $\varphi(x^n) = \varphi(x)^n$ for all $n \in \mathbb{Z}^+$.
- (b) Do part (a) for $n = -1$ and deduce that $\varphi(x^n) = \varphi(x)^n$ for all $n \in \mathbb{Z}$.

Solution.

- (a) Inducting on n , the base case of $n = 1$ is trivial. If the result holds for $n \in \mathbb{Z}^+$, then

$$\varphi(x^{n+1}) = \varphi(x^n x) = \varphi(x^n) \varphi(x) = \varphi(x)^n \varphi(x) = \varphi(x)^{n+1}.$$

- (b) Let 1_G and 1_H be the identity elements of G and H , respectively. Then for $n = 0$, we have

$$\varphi(1_G) = \varphi(1_G \cdot 1_G) = \varphi(1_G) \varphi(1_G),$$

so that $\varphi(1_G) = 1_H$. For $n = -1$ and $x \in G$, we have

$$1_H = \varphi(1_G) = \varphi(xx^{-1}) = \varphi(x) \varphi(x^{-1})$$

so that $\varphi(x^{-1}) = \varphi(x)^{-1}$. Then for $n < -1$, we have

$$\varphi(x^n) = \varphi((x^{-1})^{-n}) = \varphi(x^{-1})^{-n} = \varphi(x)^{-n} = \varphi(x)^n. \quad \blacksquare$$

Exercise 1.6.2

If $\varphi : G \rightarrow H$ is an isomorphism, prove that $|\varphi(x)| = |x|$ for all $x \in G$. Deduce that any two isomorphic groups have the same number of elements of order n for each $n \in \mathbb{Z}^+$. Is the result true if φ is only assumed to be a homomorphism?

Solution. Let $x \in G$. Let $|x| = m$ and $|\varphi(x)| = n$. Then

$$1_H = \varphi(1_G) = \varphi(x^m) = \varphi(x)^m$$

so that $n \leq m$. Similarly,

$$1_G = \varphi^{-1}(1_H) = \varphi^{-1}(\varphi(x)^n) = x^n$$

so that $m \leq n$. It follows that $m = n$, i.e., $|\varphi(x)| = |x|$. In particular, this shows that both x and $\varphi(x)$ both have either finite or infinite order. Moreover, since φ is bijective, it must be bijective between the elements of order n in G and the elements of order n in H . Hence, any two isomorphic groups have the same number of elements of order n for each $n \in \mathbb{Z}^+$.

The result is not true if φ is only assumed to be a homomorphism: the map $\varphi : G \rightarrow 1$ is a homomorphism for any group G , but G may have elements of various orders while 1 has only the identity element of order 1. $\blacksquare$

Exercise 1.6.3

If $\varphi : G \rightarrow H$ is an isomorphism, prove that G is Abelian if and only if H is Abelian. If $\varphi : G \rightarrow H$ is a homomorphism, what additional conditions on φ (if any) are sufficient to ensure that if G is Abelian, then so is H ?

Solution. Note that φ is bijective, hence φ^{-1} exists and is also an isomorphism.

($\Rightarrow$) Suppose that G is Abelian. Then for $h_1 = \varphi(g_1), h_2 = \varphi(g_2) \in H$ for some $g_1, g_2 \in G$, we have

$$h_1 h_2 = \varphi(g_1) \varphi(g_2) = \varphi(g_1 g_2) = \varphi(g_2 g_1) = \varphi(g_2) \varphi(g_1) = h_2 h_1.$$

Hence, H is Abelian.

($\Leftarrow$) Suppose that H is Abelian. Then for $g_1 = \varphi^{-1}(h_1), g_2 = \varphi^{-1}(h_2) \in G$ for some $h_1, h_2 \in H$, we have

$$g_1 g_2 = \varphi^{-1}(h_1) \varphi^{-1}(h_2) = \varphi^{-1}(h_1 h_2) = \varphi^{-1}(h_2 h_1) = \varphi^{-1}(h_2) \varphi^{-1}(h_1) = g_2 g_1.$$

Hence, G is Abelian.

Let $\varphi : G \rightarrow H$ be a homomorphism. In the ($\Rightarrow$) direction, observe that the only property of φ used was its surjectivity to guarantee the existence of g_1 and g_2 such that $\varphi(g_1) = h_1$ and $\varphi(g_2) = h_2$. Hence, if φ is surjective, then H is Abelian whenever G is Abelian. ■

Exercise 1.6.4

Prove that the multiplicative groups $\mathbb{R} - \{0\}$ and $\mathbb{C} - \{0\}$ are not isomorphic.

Solution. Note that $i \in \mathbb{C} - \{0\}$ has order 4, but there is no element in $\mathbb{R} - \{0\}$ that has order 4. ■

Exercise 1.6.5

Prove that the additive groups $\mathbb{R}$ and $\mathbb{Q}$ are not isomorphic.

Solution. $\mathbb{R}$ is uncountable, while $\mathbb{Q}$ is countable. But bijections preserve cardinality. ■

Exercise 1.6.6

Prove that the additive groups $\mathbb{Z}$ and $\mathbb{Q}$ are not isomorphic.

Solution. Every nonzero element of $\mathbb{Z}$ has infinite order. However, $1/k \in \mathbb{Q}$ has order k . ■

Exercise 1.6.7

Prove that D_8 and Q_8 are not isomorphic.

Solution. D_8 has only 1 element of order 4, while Q_8 has 3 elements of order 4. ■

Exercise 1.6.8

Prove that if $n \neq m$, then S_n and S_m are not isomorphic.

Solution. If $n \neq m$, then $n! \neq m!$ so that $|S_n| \neq |S_m|$. ■

Exercise 1.6.9

Prove that D_{24} and S_4 are not isomorphic.

Solution. D_{24} has elements of order 12, but S_4 does not. ■

Exercise 1.6.10

Fill in the details of the proof that the symmetric groups S_Δ and S_Ω are isomorphic if $|\Delta| = |\Omega|$ as follows: let $\theta : \Delta \rightarrow \Omega$ be a bijection. Define

$$\varphi : S_\Delta \rightarrow S_\Omega \quad \text{by} \quad \varphi(\sigma) = \theta \circ \sigma \circ \theta^{-1} \quad \text{for all } \sigma \in S_\Delta$$

and prove the following:

- (a) Prove that φ is well-defined, that is, if σ is a permutation of Δ then $\theta \circ \sigma \circ \theta^{-1}$ is a permutation of Ω .
- (b) Prove that φ is a bijection from S_Δ onto S_Ω by finding a two-sided inverse for φ .
- (c) Prove that φ is a homomorphism, that is, $\varphi(\sigma \circ \tau) = \varphi(\sigma) \circ \varphi(\tau)$.

REMARK. Note the similarity to the change of basis or similarity transformations for matrices (we shall see the connections between these later in the text).

Solution.

- (a) Since $\theta : \Delta \rightarrow \Omega$ and $\theta^{-1} : \Omega \rightarrow \Delta$ are both bijections with σ a bijection, it follows their composition is a bijection. Moreover, $\theta \circ \sigma \circ \theta^{-1}$ is a permutation of Ω , so that φ is well-defined.
- (b) Consider the mapping

$$\psi : S_\Omega \rightarrow S_\Delta \quad \text{by} \quad \psi(\tau) = \theta^{-1} \circ \tau \circ \theta \quad \text{for all } \tau \in S_\Omega$$

By the above discussion, ψ is well-defined. For any $\sigma \in S_\Delta$, we have

$$\psi(\varphi(\sigma)) = \theta^{-1} \circ (\theta \circ \sigma \circ \theta^{-1}) \circ \theta = \text{id}_{S_\Delta} \circ \sigma \circ \text{id}_{S_\Delta} = \sigma$$

and for any $\tau \in S_\Omega$, we have

$$\varphi(\psi(\tau)) = \theta \circ (\theta^{-1} \circ \tau \circ \theta) \circ \theta^{-1} = \text{id}_{S_\Omega} \circ \tau \circ \text{id}_{S_\Omega} = \tau$$

so that ψ is a two-sided inverse for φ . Hence, φ is a bijection from S_Δ onto S_Ω .

- (c) Let $\sigma, \tau \in S_\Delta$. Then

$$\varphi(\sigma \circ \tau) = \theta \circ (\sigma \circ \tau) \circ \theta^{-1} = (\theta \circ \sigma) \circ (\tau \circ \theta^{-1}) = (\theta \circ \sigma \circ \theta^{-1}) \circ (\theta \circ \tau \circ \theta^{-1}) = \varphi(\sigma) \circ \varphi(\tau)$$

so that φ is a homomorphism. By the above, φ is an isomorphism, and hence $S_\Delta \cong S_\Omega$. ■

Exercise 1.6.11

Let A and B be groups. Prove that $A \times B \cong B \times A$.

Solution. The mapping

$$\varphi : A \times B \rightarrow B \times A \quad \text{given by} \quad (a, b) \mapsto (b, a)$$

for all $a \in A$ and $b \in B$ is clearly bijective. Moreover, for any $(a_1, b_1), (a_2, b_2) \in A \times B$, we have

$$\varphi((a_1, b_1)(a_2, b_2)) = \varphi((a_1 a_2, b_1 b_2)) = (b_1 b_2, a_1 a_2) = (b_1, a_1)(b_2, a_2) = \varphi(a_1, b_1)\varphi(a_2, b_2)$$

so that φ is a homomorphism. Therefore, φ is an isomorphism and $A \times B \cong B \times A$. ■

Exercise 1.6.12

Let A , B , and C be groups and let $G = A \times B$ and $H = B \times C$. Prove that $G \times C \cong A \times H$.

Solution. Consider the mapping

$$\varphi : G \times C \rightarrow A \times H \quad \text{by} \quad ((a, b), c) \mapsto (a, (b, c))$$

for all $a \in A$, $b \in B$, and $c \in C$. The proof is similar to the previous exercise, hence we omit the details to conclude that φ is an isomorphism and $G \times C \cong A \times H$. ■

Exercise 1.6.13

Let G and H be groups and let $\varphi : G \rightarrow H$ be a homomorphism. Prove that the image of φ , $\varphi(G)$, is a **subgroup** of H . Prove that if φ is injective then $G \cong \varphi(G)$.

Solution. Observe that $1_H \in \varphi(G)$ since $\varphi(1_G) = 1_H$, so that $\varphi(G)$ is nonempty. To show closure, let $h_1, h_2 \in \varphi(G)$. Then there exist $g_1, g_2 \in G$ such that $\varphi(g_1) = h_1$ and $\varphi(g_2) = h_2$. Then

$$h_1 h_2 = \varphi(g_1) \varphi(g_2) = \varphi(g_1 g_2) \in \varphi(G)$$

so that $h_1 h_2 \in \varphi(G)$. Finally,

$$h_1^{-1} = \varphi(g_1)^{-1} = \varphi(g_1^{-1}) \in \varphi(G)$$

so that $h_1^{-1} \in \varphi(G)$. Therefore, $\varphi(G)$ is a subgroup of H .

Suppose φ is injective. Then the mapping

$$\psi : G \rightarrow \varphi(G) \quad \text{by} \quad g \mapsto \varphi(g)$$

is well-defined. For any $g_1, g_2 \in G$,

$$\psi(g_1 g_2) = \varphi(g_1 g_2) = \varphi(g_1) \varphi(g_2) = \psi(g_1) \psi(g_2)$$

so that ψ is a homomorphism. Moreover, if $\psi(g_1) = \psi(g_2)$, then $\varphi(g_1) = \varphi(g_2)$, and since φ is injective, $g_1 = g_2$. Hence, ψ is injective. For any $h \in \varphi(G)$, there exists $g \in G$ such that $\varphi(g) = h$, and hence $\psi(g) = h$. Therefore, ψ is surjective, hence ψ is an isomorphism and $G \cong \varphi(G)$. ■

Exercise 1.6.14

Let G and H be groups and let $\varphi : G \rightarrow H$ be a homomorphism. Define the **kernel** of φ to be $\{g \in G \mid \varphi(g) = 1_H\}$ (so the kernel is the set of elements in G which map to the identity of H , i.e., is the fiber over the identity of H). Prove that the kernel of φ is a subgroup of G . Prove that φ is injective if and only if the kernel of φ is the identity subgroup of G .

Solution. Denote the kernel of φ by $\ker \varphi$. Since $\varphi(1_G) = 1_H$, then $1_G \in \ker \varphi$, so that $\ker \varphi$ is nonempty. To show closure, let $g_1, g_2 \in \ker \varphi$. Then

$$\varphi(g_1 g_2) = \varphi(g_1) \varphi(g_2) = 1_H 1_H = 1_H$$

so that $g_1 g_2 \in \ker \varphi$. Finally,

$$\varphi(g_1^{-1}) = \varphi(g_1)^{-1} = 1_H^{-1} = 1_H$$

so that $g_1^{-1} \in \ker \varphi$. Therefore, $\ker \varphi$ is a subgroup of G .

- ($\Rightarrow$) If φ is injective, then for $g \in \ker \varphi$, we have $\varphi(g) = 1_H = \varphi(1_G)$. Then $g = 1_G$, so that $\ker \varphi = \{1_G\}$.
 ($\Leftarrow$) If $\ker \varphi = \{1_G\}$, then for $g_1, g_2 \in G$, if $\varphi(g_1) = \varphi(g_2)$, we have $\varphi(g_1)\varphi(g_2)^{-1} = 1_H$, so $g_1g_2^{-1} \in \ker \varphi$. Hence, $g_1g_2^{-1} = 1_G$, and thus $g_1 = g_2$. Therefore, φ is injective. ■

Exercise 1.6.15

Define a map $\pi : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $\pi((x, y)) = x$. Prove that π is a homomorphism and find the kernel of π .

Solution. Let $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$. Then

$$\pi((x_1, y_1) + (x_2, y_2)) = \pi((x_1 + x_2, y_1 + y_2)) = x_1 + x_2 = \pi((x_1, y_1)) + \pi((x_2, y_2))$$

so that π is a homomorphism. Moreover, if $(x, y) \in \ker(\pi)$, then $\pi((x, y)) = x = 0$. Hence,

$$\ker(\pi) = \{(0, y) \mid y \in \mathbb{R}\}$$

Exercise 1.6.16

Let A and B be groups and let G be their direct product, $A \times B$. Prove that the maps $\pi_1 : G \rightarrow A$ and $\pi_2 : G \rightarrow B$ defined by $\pi_1((a, b)) = a$ and $\pi_2((a, b)) = b$ are homomorphisms and find their kernels.

Solution. The proof is similar to the previous exercise, hence we omit the details to conclude that π_1 and π_2 are homomorphisms. Moreover, if $(a, b) \in \ker(\pi_1)$, then $\pi_1((a, b)) = a = 1_A$. If $(a, b) \in \ker(\pi_2)$, then $\pi_2((a, b)) = b = 1_B$. Hence,

$$\ker(\pi_1) = \{(1_A, b) \mid b \in B\}, \quad \text{and} \quad \ker(\pi_2) = \{(a, 1_B) \mid a \in A\}$$

Exercise 1.6.17

Let G be any group. Prove that the map from G to itself defined by $g \mapsto g^{-1}$ is a homomorphism if and only if G is Abelian.

Solution.

($\Rightarrow$) Suppose φ is a homomorphism. Then for $g, h \in G$, we have

$$(gh)^{-1} = \varphi(gh) = \varphi(g)\varphi(h) = g^{-1}h^{-1} = h^{-1}g^{-1}.$$

Since inverses are unique, then $hg = gh$ so that G is Abelian.

($\Leftarrow$) Suppose G is Abelian. Then for $g, h \in G$, we have

$$\varphi(gh) = (gh)^{-1} = h^{-1}g^{-1} = g^{-1}h^{-1} = \varphi(g)\varphi(h),$$

so that φ is a homomorphism. ■

Exercise 1.6.18

Let G be any group. Prove that the map from G to itself defined by $g \mapsto g^2$ is a homomorphism if and only if G is Abelian.

Solution. The proof is similar to [Exercise 1.6.17](#), hence we omit the details. ■

Exercise 1.6.19

Let $G = \{z \in \mathbb{C} \mid z^n = 1 \text{ for some } n \in \mathbb{Z}^+\}$. Prove that for any fixed integer $k > 1$ the map from G to itself defined by $z \mapsto z^k$ is a surjective homomorphism but is not an isomorphism.

Solution. Since $G \subseteq \mathbb{C}$, then G is Abelian. Let $\varphi : z \rightarrow z^k$. Hence, for any $z, w \in G$, we have

$$\varphi(zw) = (zw)^k = z^k w^k = \varphi(z)\varphi(w)$$

so that φ is a homomorphism. To show surjectivity, let $w \in G$ such that $w^n = 1$ for some $n \in \mathbb{Z}^+$. Note that $w^{1/k} \in G$ because $(z^{1/k})^{nk} = z^n = 1$. Then

$$\varphi(z^{1/k}) = (z^{1/k})^k = z$$

so that φ is surjective. Finally, any $z = e^{2\pi i/k} \in G$ is such that $\varphi(z) = z^k = 1$ with $z \neq 1$ since $k > 1$. Hence, φ is not injective and thus not an isomorphism. ■

Exercise 1.6.20

Let G be a group and let $\text{Aut}(G)$ be the set of all isomorphisms from G onto G . Prove that $\text{Aut}(G)$ is a group under function composition (called the **automorphism group** of G and the elements of $\text{Aut}(G)$ are called **automorphisms** of G).

Solution. Observe that $\text{Aut}(G)$ is associative under function composition and has identity through the identity map $\text{id} : G \rightarrow G$ given by $\text{id}(g) = g$. Moreover, $\varphi \circ \psi$ for $\varphi, \psi \in \text{Aut}(G)$ is a bijection. Then for any $g, h \in G$, we have

$$(\varphi \circ \psi)(gh) = \varphi(\psi(gh)) = \varphi(\psi(g)\psi(h)) = \varphi(\psi(g))\varphi(\psi(h)) = (\varphi \circ \psi)(g)(\varphi \circ \psi)(h),$$

so that $\varphi \circ \psi$ is a homomorphism. Hence, $\varphi \circ \psi \in \text{Aut}(G)$, and $\text{Aut}(G)$ is closed under function composition. Finally, for any $\varphi \in \text{Aut}(G)$, then φ^{-1} exists and is a bijection. Then

$$\varphi^{-1}(gh) = \varphi^{-1}(\varphi(\varphi^{-1}(g))\varphi(\varphi^{-1}(h))) = \varphi^{-1}(\varphi(\varphi^{-1}(g)\varphi^{-1}(h))) = \varphi^{-1}(g)\varphi^{-1}(h),$$

so that φ^{-1} is a homomorphism. Hence, $\varphi^{-1} \in \text{Aut}(G)$, and $(\text{Aut}(G), \circ)$ is a group. ■

Exercise 1.6.21

Prove that for each fixed nonzero $k \in \mathbb{Q}$ the map from $\mathbb{Q}$ to itself defined by $q \mapsto kq$ is an automorphism of $\mathbb{Q}$.

Solution. Let $\varphi : \mathbb{Q} \rightarrow \mathbb{Q}$ be defined by $\varphi(q) = kq$ for all $q \in \mathbb{Q}$. For any $p, q \in \mathbb{Q}$, we have

$$\varphi(p + q) = k(p + q) = kp + kq = \varphi(p) + \varphi(q),$$

so that φ is a homomorphism. Bijectivity is clear, hence $\varphi \in \text{Aut}(\mathbb{Q})$. ■

Exercise 1.6.22

Let A be an Abelian group and fix some $k \in \mathbb{Z}$. Prove that the map $a \mapsto a^k$ is a homomorphism from A to itself. If $k = -1$ prove that this homomorphism is an isomorphism (i.e., is an automorphism of A).

Solution. Let $\varphi : A \rightarrow A$ be defined by $\varphi(a) = a^k$ for all $a \in A$. For any $a_1, a_2 \in A$, we have

$$\varphi(a_1 a_2) = (a_1 a_2)^k = a_1^k a_2^k = \varphi(a_1) \varphi(a_2)$$

so that φ is a homomorphism. For $k = -1$, see [Exercise 1.6.17](#) to conclude that $\varphi \in \text{Aut}(A)$. ■

Exercise 1.6.23

Let G be a finite group which possesses an automorphism σ such that $\sigma(g) = g$ if and only if $g = 1$. If σ^2 is the identity map from G to G , prove that G is Abelian (such an automorphism σ is called **fixed point free** of order 2).

HINT. Show that every element of G can be written in the form $x^{-1}\sigma(x)$ and apply σ to such an expression.

Solution. Define the map

$$f : G \rightarrow G \quad \text{by} \quad x \mapsto x^{-1}\sigma(x)$$

for all $x \in G$. If $f(x) = f(y)$ for $x, y \in G$, then $x^{-1}\sigma(x) = y^{-1}\sigma(y)$ so that $yx^{-1} = \sigma(yx^{-1})$. Since σ is fixed point free, then $yx^{-1} = 1$ and $x = y$. Hence, f is injective. Since G is finite, then f is also surjective, and every element of G may be written in the form $x^{-1}\sigma(x)$ for some $x \in G$.

Let $g \in G$ be arbitrary. Then there exists $x \in G$ such that $g = x^{-1}\sigma(x)$. Applying σ to both sides, we have

$$\sigma(g) = \sigma(x^{-1}\sigma(x)) = \sigma(x^{-1})\sigma^2(x) = \sigma(x)^{-1}x = (x^{-1}\sigma(x))^{-1} = g^{-1}.$$

Then for any $g, h \in G$, we have

$$\sigma(gh) = (gh)^{-1} = h^{-1}g^{-1} = \sigma(h)\sigma(g) = \sigma(hg).$$

Since σ is injective, then $gh = hg$ and G is Abelian. ■

Exercise 1.6.24

Let G be a finite group and let x and y be distinct elements of order 2 in G that generate G . Prove that $G \cong D_{2n}$, where $n = |xy|$.

Solution. To show that G is isomorphic to D_{2n} , we first need to produce an element $t \in G$ of order n along with an element $x \in G$ of order 2 such that $xt = t^{-1}x$. The clear choice is to let $t = xy$ and keep x as is. Note that $|t| = |xy| = n$ by assumption, and $|x| = 2$ by assumption. Moreover,

$$xt = x(xy) = (xx)y = y = (xy)^{-1}x = t^{-1}x$$

so that the elements $x, t \in G$ satisfy the relations of D_{2n} . Additionally, observe that $y = xt$ so that t and x generate G . Following this, it is easy to see that every element of G can be written in the form $x^i t^j$ for $i \in \{0, 1\}$ and $0 \leq j < n$. For $x = 0$, we have the elements in the cyclic subgroup generated by t , and for $x = 1$, we have the elements $\{x, xt, xt^2, \dots, xt^{n-1}\}$. Since $|t| = n$, then these are all distinct elements of G . Hence, $|G| = 2n$.

Since the relations of D_{2n} are satisfied by $x, t \in G$, G is generated by x and t , and $|G| = |D_{2n}| = 2n$, it follows that the map

$$\varphi : D_{2n} \rightarrow G \quad \text{by} \quad r \mapsto t, s \mapsto x$$

extends to an isomorphism from D_{2n} onto G . Therefore, $G \cong D_{2n}$. ■

Exercise 1.6.25

Let $n \in \mathbb{Z}^+$, let r and s be the usual generators of D_{2n} and let $\theta = 2\pi/n$.

(a) Prove that the matrix

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

is the matrix of the linear transformation which rotates the x, y -plane about the origin in a counterclockwise direction by θ radians.

(b) Prove that the map $\varphi : D_{2n} \rightarrow \text{GL}_2(\mathbb{R})$ defined on generators by

$$\varphi(r) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \quad \text{and} \quad \varphi(s) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

extends to a homomorphism of D_{2n} into $\text{GL}_2(\mathbb{R})$.

(c) Prove that the homomorphism φ defined above is injective.

Solution.

(a) Let $\mathbb{R}^2$ be generated by the standard basis vectors $\mathbf{e}_1 = (1 \ 0)^T$ and $\mathbf{e}_2 = (0 \ 1)^T$. Let T denote the linear transformation which rotates the x, y -plane about the origin in a counterclockwise direction by θ radians. Then

$$T(\mathbf{e}_1) = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}, \quad \text{and} \quad T(\mathbf{e}_2) = \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix}.$$

Then the matrix of T with respect to the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ is exactly the desired matrix.

(b) It is easy to show by induction that for any positive k , then $\varphi(r)^k$ is given by the following matrix:

$$\varphi(r)^k = \begin{pmatrix} \cos(k\theta) & -\sin(k\theta) \\ \sin(k\theta) & \cos(k\theta) \end{pmatrix}$$

so that $\varphi(r)^n = I$. It is fairly straightforward to see that the set of vectors $\{T(\mathbf{e}_1), T(\mathbf{e}_2)\}$ is orthonormal, so $\varphi(r)$ is an orthogonal matrix, and hence $\varphi(r)^{-1} = \varphi(r)^T$. Moreover, we can calculate that $\varphi(s)^2 = I$. Lastly, we calculate that

$$\varphi(s)\varphi(r) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} = \begin{pmatrix} \sin \theta & \cos \theta \\ \cos \theta & -\sin \theta \end{pmatrix} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \varphi(r)^{-1}\varphi(s)$$

so that $\varphi(s)\varphi(r) = \varphi(r)^{-1}\varphi(s)$. Since these relations match those of D_{2n} , then φ extends to a homomorphism from D_{2n} to $\text{GL}_2(\mathbb{R})$.

(c) To show that φ is injective, we show that $\ker \varphi$ is trivial. Let $g \in \ker \varphi$.

- If $g = r^k$ for some $0 \leq k < n$, then $\varphi(r^k) \neq I$, since $\varphi(r)^k$ is a rotation matrix and only $\varphi(r)^n = I$.
- If $g = sr^k$ for some $0 \leq k < n$, then $\varphi(sr^k) = \varphi(s)\varphi(r)^k$. Note that $\varphi(s)$ has determinant -1 while $\varphi(r)^k$ has determinant 1 since it is a rotation matrix. Hence, $\varphi(sr^k)$ has determinant -1 and cannot be the identity matrix.

Therefore, the only element in $\ker \varphi$ is the identity element of D_{2n} , so that φ is injective. ■

Exercise 1.6.26

Let i and j be the generators of Q_8 described in Section 5. Prove that the map φ from Q_8 to $GL_2(\mathbb{C})$ defined on generators by

$$\varphi(i) = \begin{pmatrix} \sqrt{-1} & 0 \\ 0 & -\sqrt{-1} \end{pmatrix} \quad \text{and} \quad \varphi(j) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

extends to a homomorphism. Prove that φ is injective.

Solution. Matrix multiplication shows that the relations described in [Exercise 1.5.3](#) are satisfied, so that φ extends to a homomorphism from Q_8 to $GL_2(\mathbb{C})$. Moreover, φ is injective since the only element that maps to the identity matrix is $1 \in Q_8$. Therefore, φ is an injective homomorphism. ■

1.7 Group Actions

Exercise 1.7.1

Let F be a field. Show that the multiplicative group of nonzero elements of F (denoted by $F^\times$) acts on the set F by $g \cdot a = ga$, where $g \in F^\times$, $a \in F$ and ga is the usual product in F of the two field elements (state clearly which axioms in the definition of a field are used).

Solution. Let $g, h \in F^\times$ and $f \in F$. Then

$$g \cdot (h \cdot f) = g \cdot (hf) = g(hf) = (gh)f = (gh) \cdot f$$

where the associativity of multiplication in F was used to conclude that $g(hf) = (gh)f$. Moreover, $1 \cdot f = 1f = f$ where 1 is the multiplicative identity in F . Therefore, the group action axioms are satisfied and $F^\times$ acts on F . ■

Exercise 1.7.2

Show that the additive group $\mathbb{Z}$ acts on itself by $z \cdot a = z + a$ for all $z, a \in \mathbb{Z}$.

Solution. Let $z, w \in \mathbb{Z}$ and $a \in \mathbb{Z}$. Then

$$z \cdot (w \cdot a) = z \cdot (w + a) = z + (w + a) = (z + w) + a = (z + w) \cdot a$$

Moreover, $0 \cdot a = 0 + a = a$ so that the group action axioms are satisfied and $\mathbb{Z}$ acts on itself. ■

Exercise 1.7.3

Show that the additive group $\mathbb{R}$ acts on the x, y plane $\mathbb{R} \times \mathbb{R}$ by $r \cdot (x, y) = (x + r, y)$.

Solution. Let $r, s \in \mathbb{R}$ and $(x, y) \in \mathbb{R}^2$. Then

$$r \cdot (s \cdot (x, y)) = r \cdot (x + s, y) = (x + s + r, y) = (x + (r + s), y) = (r + s) \cdot (x, y)$$

Moreover, $0 \cdot (x, y) = (x + 0, y) = (x, y)$ so that the group action axioms are satisfied and $\mathbb{R}$ acts on $\mathbb{R}^2$. ■

Exercise 1.7.4

[1.7.4] Let G be a group acting on a set A and fix some $a \in A$. Show that the following sets are subgroups of G (cf. [Exercise 1.1.26](#)):

- (a) the kernel of the action,
- (b) $\{g \in G \mid ga = a\}$ —this subgroup is called the **stabilizer** of a in G .

Solution.

- (a) Suppose g, h are in the kernel of the action. Then for any $a \in A$, we have

$$gh \cdot a = g \cdot (h \cdot a) = g \cdot a = a$$

so that gh is in the kernel of the action. Also, for any g in the kernel of the action and any $a \in A$, we have

$$g^{-1} \cdot a = g^{-1} \cdot (g \cdot a) = (g^{-1}g) \cdot a = 1 \cdot a = a$$

so that g^{-1} is in the kernel of the action. Therefore, the kernel of the action is a subgroup of G .

(b) Let G_a denote the stabilizer of a in G . Suppose $g, h \in G_a$. Then

$$gh \cdot a = g \cdot (h \cdot a) = g \cdot a = a$$

so that $gh \in G_a$. Also, for any $g \in G_a$, we have

$$g^{-1} \cdot a = g^{-1} \cdot (g \cdot a) = (g^{-1}g) \cdot a = 1 \cdot a = a$$

so that $g^{-1} \in G_a$. Therefore, G_a is a subgroup of G . ■

Exercise 1.7.5

Prove that the kernel of an action of the group G on the set A is the same as the kernel of the corresponding permutation representation $G \rightarrow S_A$ (cf. Exercise 1.6.14).

Solution. Let K denote the kernel of the action, and let K' denote $\ker \varphi$, where $\varphi : G \rightarrow S_A$ is the permutation representation corresponding to the action. Suppose $g \in K$. Then $g \cdot a = a$ for every $a \in A$, hence $\varphi(g)$ is the identity permutation in S_A and $g \in K'$. Conversely, suppose $g \in K'$. Then $\varphi(g)$ is the identity permutation in S_A , so that for every $a \in A$, we have $\varphi(g)(a) = a$. But $\varphi(g)(a) = g \cdot a$ by definition of the permutation representation, so that $g \cdot a = a$ for every $a \in A$ and $g \in K$. Therefore, $K = K'$. ■

Exercise 1.7.6

Prove that a group G acts faithfully on a set A if and only if the kernel of the action is the set consisting only of the identity.

Solution.

($\Rightarrow$) Suppose G acts faithfully on A with kernel K . Certainly, $1 \cdot a = a$ for all $a \in A$, hence $1 \in K$. If $g \in K$, then $g \cdot a = a$. Faithfulness of the action implies that $g = 1$, so that $K = \{1\}$.

($\Leftarrow$) Suppose the kernel of the action is the set consisting only of the identity. Let $g, h \in G$ be such that $g \cdot a = h \cdot a$ for every $a \in A$. Then

$$h^{-1}g \cdot a = h^{-1} \cdot (g \cdot a) = h^{-1} \cdot (h \cdot a) = (h^{-1}h) \cdot a = 1 \cdot a = a$$

so that $h^{-1}g$ is in the kernel of the action. By assumption, then $h^{-1}g$ is the identity element of G , and $g = h$. Therefore, the action is faithful. ■

Exercise 1.7.7

Prove that in Example 2 in this section the action is faithful.

Solution. Let V be a vector space over a field F . Let $F^\times$ act on V by $r \cdot v = rv$ for all $r \in F^\times$ and $v \in V$. Suppose $r \in F^\times$ is in the kernel of the action. Then for every $v \in V$, we have $r \cdot v = v$, so that $rv = v$. Since V is a vector space, then it contains the nonzero vector $v = 1$. Hence, $r(1) = 1$ so that $r = 1$. Therefore, the kernel of the action is the set consisting only of the identity, and by the previous exercise, the action is faithful. ■

Exercise 1.7.8

Let A be a nonempty set and let k be a positive integer with $k \leq |A|$. The symmetric group S_A acts on the set B consisting of all subsets of A of cardinality k by

$$\sigma(\{a_1, \dots, a_k\}) = \{\sigma(a_1), \dots, \sigma(a_k)\}.$$

- (a) Prove that this is a group action.
 (b) Describe explicitly how the elements $(1\ 2)$ and $(1\ 2\ 3)$ act on the six 2-element subsets of $\{1, 2, 3, 4\}$.

Solution.

- (a) Let $\sigma, \tau \in S_A$. Then for any cardinality k subset $\{x_1, \dots, x_k\}$ of A , we have

$$\begin{aligned} \sigma \cdot (\tau \cdot \{x_1, \dots, x_k\}) &= \sigma \cdot \{\tau(x_1), \dots, \tau(x_k)\} \\ &= \{\sigma(\tau(x_1)), \dots, \sigma(\tau(x_k))\} \\ &= (\sigma \circ \tau) \cdot \{x_1, \dots, x_k\} \end{aligned}$$

Moreover, $1 \cdot \{x_1, \dots, x_k\} = \{1(x_1), \dots, 1(x_k)\} = \{x_1, \dots, x_k\}$. Then it is a group action.

- (b)

A	$(1\ 2) \cdot A$	$(1\ 2\ 3) \cdot A$
$\{1, 2\}$	$\{2, 1\}$	$\{2, 3\}$
$\{1, 3\}$	$\{2, 3\}$	$\{2, 1\}$
$\{1, 4\}$	$\{2, 4\}$	$\{2, 4\}$
$\{2, 3\}$	$\{1, 3\}$	$\{3, 1\}$
$\{2, 4\}$	$\{1, 4\}$	$\{3, 4\}$
$\{3, 4\}$	$\{3, 4\}$	$\{1, 4\}$

■

Exercise 1.7.9

Do both parts of the preceding exercise with “ordered k -tuples” in place of “ k -element subsets,” where the action on k -tuples is defined as above but with set braces replaced by parentheses.

REMARK. Note that, for example, the 2-tuples $(1, 2)$ and $(2, 1)$ are different even though the sets $\{1, 2\}$ and $\{2, 1\}$ are the same, so the sets being acted upon are different.

Solution. The work is similar as above, except there are 12 2-tuples as $(1, 2) \neq (2, 1)$. ■

Exercise 1.7.10

With reference to the preceding two exercises determine:

- (a) for which values of k the action of S_n on k -element subsets is faithful, and
 (b) for which values of k the action of S_n on ordered k -tuples is faithful.

Solution.

- (a) If $k = |A|$, then any subset S of A with cardinality k is equal to A . Then any $\sigma \in S_n$ will fix S , so the action is not faithful.

Let $\sigma \in S_n$ be a non-identity permutation. Then there exists some $a \in A$ such that $\sigma(a) \neq a$. If $k < |A|$, then we can choose a k -element subset S of A such that $a \in S$ and $\sigma(a) \notin S$. Then $\sigma(S) \neq S$, hence no non-identity permutation fixes every k -element subset of A . Therefore, the action is faithful for all $1 \leq k < |A|$.

- (b) Let $\sigma \in S_n$ be a non-identity permutation. Then there exists some $a \in A$ such that $\sigma(a) \neq a$. Consider the ordered k -tuple $T = (a, x_2, x_3, \dots, x_k)$ where $x_2, x_3, \dots, x_k$ are distinct elements of A different from a and $\sigma(a)$. Then $\sigma(T) = (\sigma(a), \sigma(x_2), \sigma(x_3), \dots, \sigma(x_k)) \neq T$ because the first element is different. Hence, no non-identity permutation fixes every ordered k -tuple of elements of A . Therefore, the action is faithful for all $1 \leq k \leq |A|$. ■

Exercise 1.7.11

Write out the cycle decomposition of the eight permutations in S_4 corresponding to the elements of D_8 given by the action of D_8 on the vertices of a square (where the vertices of the square are labeled as in Section 2).

Solution. Let $\varphi : D_8 \rightarrow S_4$ be the permutation representation of the action of D_8 on the vertices of a square $\{1, 2, 3, 4\}$, where the vertices are labeled in order around the square starting in the top left corner, then going clockwise. Then the cycle decompositions are as follows:

$$\begin{aligned} \varphi(1) &= 1 & \varphi(r) &= (1\ 2\ 3\ 4) \\ \varphi(r^2) &= (1\ 3)(2\ 4) & \varphi(r^3) &= (1\ 4\ 3\ 2) \\ \varphi(s) &= (2\ 4) & \varphi(sr) &= (1\ 4)(2\ 3) \\ \varphi(sr^2) &= (1\ 3) & \varphi(sr^3) &= (1\ 2)(3\ 4). \end{aligned}$$

Exercise 1.7.12

Assume n is an even positive integer and show that D_{2n} acts on the set consisting of pairs of opposite vertices of a regular n -gon. Find the kernel of this action (label vertices as usual).

Solution. Let the n -gon have the usual labeling of vertices. Put P_k as the pair of opposite vertices

$$P_k = \{k, k + n/2\} \quad \text{defined for all } 1 \leq k \leq n/2,$$

with P denoting the set of all opposite pairs of vertices. Let D_{2n} act on P by

$$g \cdot P_i = g(P_i) = P_j$$

where P_j is the pair of opposite vertices containing the images of the vertices in P_i under the action of g on the vertices of the n -gon. Since elements of D_{2n} are symmetries of the n -gon, then they will map pairs of opposite vertices to other pairs of opposite vertices. Hence, D_{2n} acts on P .

Let $g \in D_{2n}$ be in the kernel of this action. Then $g \cdot P_i = P_i$ for every $P_i \in P$. Consider some $P_i = \{i, i + n/2\} \in P$. If g maps i to $i + n/2$, then it must map $i + n/2$ to i , which corresponds to the rotation $r^{n/2}$. If g maps i to i , then it must map $i + n/2$ to $i + n/2$, which corresponds to the identity element.

Suppose g is a reflection. If the reflection is about the axis through the vertices in P_i , then $i + 1$ is mapped to $i - 1$. Then either $i - 1 = i + 1$ or $i - 1 = i + 1 + n/2$, which is impossible. If the reflection is about the axis through the midpoints of the edges between i and $i + 1$ and between $i + n/2$ and $i + n/2 + 1$, then i is mapped to $i + 1$. Then either $i + 1 = i$ or $i + 1 = i + n/2$, which is impossible. Therefore, no reflection can be in the kernel of this action. ■

Exercise 1.7.13

Find the kernel of the left regular action.

Solution. If g is in the kernel of the left regular action of G on itself, then $g \cdot 1 = 1$ in particular. Then $g1 = 1$, implying that $g = 1$. Then the kernel of the left regular action is trivial. ■

Exercise 1.7.14

Let G be a group and let $A = G$. Show that if G is non-Abelian then the maps defined by

$$g \cdot a = ag \quad \text{for all } g, a \in G$$

do not satisfy the axioms of a (left) group action of G on itself.

Solution. Since G is non-Abelian, there exists $g, h \in G$ such that $gh \neq hg$. Assume, by way of contradiction, that the above maps do satisfy the axioms of a left group action. Then for any $a \in A$, we have

$$g \cdot (h \cdot a) = g \cdot (ah) = (ah)g = a(hg)$$

On the other hand, $(gh) \cdot a = a(gh)$. Since $gh \neq hg$, then $a(hg) \neq a(gh)$ for some $a \in A$. This contradicts the assumption that the above maps satisfy the axioms of a left group action. Therefore, the maps do not satisfy the axioms of a left group action. ■

Exercise 1.7.15

Let G be any group and let $A = G$. Show that the maps defined by

$$g \cdot a = ag^{-1} \quad \text{for all } g, a \in G$$

do satisfy the axioms of a (left) group action of G on itself.

Solution. Let $g, h \in G$. Then for any $a \in A$, we have

$$g \cdot (h \cdot a) = g \cdot (ah^{-1}) = a(h^{-1}g^{-1}) = a(gh)^{-1} = (gh) \cdot a$$

Moreover, $1 \cdot a = a1^{-1} = a1 = a$. Then the maps define a left group action. ■

Exercise 1.7.16

[1.7.16] Let G be any group and let $A = G$. Show that the maps defined by

$$g \cdot a = gag^{-1} \quad \text{for all } g, a \in G$$

do satisfy the axioms of a (left) group action (this action of G on itself is called *conjugation*).

Solution. Let $g, h \in G$. Then for any $a \in A$, we have

$$g \cdot (h \cdot a) = g \cdot (hah^{-1}) = g(hah^{-1})g^{-1} = (gh)a(gh)^{-1} = (gh) \cdot a$$

Moreover, $1 \cdot a = 1a1^{-1} = a$. Then the maps define a left group action. ■

Exercise 1.7.17

Let G be a group and let G act on itself by left conjugation, so each $g \in G$ maps G to G by

$$x \mapsto gxg^{-1}.$$

For fixed $g \in G$, prove that conjugation by g is an isomorphism from G onto itself (i.e., an automorphism of G). Deduce that x and gxg^{-1} have the same order for all $x \in G$ and that for any subset A of G , $|A| = |gAg^{-1}|$ (here $gAg^{-1} = \{gag^{-1} \mid a \in A\}$).

Solution. We prove that the associated permutation $\varphi : G \rightarrow G$ defined by $\varphi(x) = gxg^{-1}$ is an isomorphism. Let $x, y \in G$. Then

$$\varphi(xy) = g(xy)g^{-1} = (gxg^{-1})(gyg^{-1}) = \varphi(x)\varphi(y)$$

so that φ is a homomorphism. To show that φ is bijective, we show that it is injective. Suppose $\varphi(x) = \varphi(y)$ for some $x, y \in G$. Then

$$gxg^{-1} = gyg^{-1}$$

implying that $x = y$. Surjectivity is clear from conjugation by g^{-1} , so φ is a bijection. Therefore, φ is an isomorphism from G onto itself.

From **Exercise 1.6.2**, we know that isomorphisms preserve order, so $|x| = |gxg^{-1}|$ for all $x \in G$. If we consider the restriction of φ to a subset A of G , then $\varphi|_A : A \rightarrow gAg^{-1}$ remains a bijection so that $|A| = |gAg^{-1}|$. ■

Exercise 1.7.18

[1.7.18] Let H be a group acting on a set A . Prove that the relation $\sim$ on A defined by

$$a \sim b \quad \text{if and only if} \quad a = hb \text{ for some } h \in H$$

is an equivalence relation. (For each $x \in A$ the equivalence class of x under $\sim$ is called the **orbit** of x under the action of H . The orbits under the action of H partition the set A .)

Solution.

- Let $a \in A$. Then $a = 1a$, hence $a \sim a$.
- Let $a, b \in A$ such that $a \sim b$. Then $a = bh$ for some $h \in H$. Then $b = h^{-1}a$, hence $b \sim a$.
- Let $a, b, c \in A$ such that $a \sim b$ and $b \sim c$. Then $a = bh_1$ and $b = ch_2$ for some $h_1, h_2 \in H$. Then $a = (ch_2)h_1 = c(h_2h_1)$, hence $a \sim c$. ■

Exercise 1.7.19

[1.7.19] Let H be a subgroup of the finite group G and let H act on G (here $A = G$) by left multiplication. Let $x \in G$ and let O be the orbit of x under this action of H . Prove that the map

$$H \rightarrow O, \quad h \mapsto hx$$

is a bijection (hence all orbits have cardinality $|H|$). From this and the preceding exercise deduce **Lagrange's Theorem**:

If G is a finite group and H is a subgroup of G then $|H|$ divides $|G|$.

Solution. Let $\varphi : H \rightarrow O$ be the map defined by $\varphi(h) = hx$ for all $h \in H$. We show that φ is a bijection. To show injectivity, suppose $\varphi(h_1) = \varphi(h_2)$ for some $h_1, h_2 \in H$. Then

$$h_1x = h_2x$$

implying that $h_1 = h_2$. To show surjectivity, let $y \in O$. Then by definition of the orbit, there exists some $h \in H$ such that $y = hx$. Hence, $\varphi(h) = y$. Therefore, φ is a bijection and $|O| = |H|$.

To deduce Lagrange's Theorem, note by the previous exercise that the orbits of the action of H on G partition G . Since each orbit has cardinality $|H|$, then $|G|$ is a sum of multiples of $|H|$. Therefore, $|H|$ divides $|G|$. ■

Exercise 1.7.20

Show that the group of rigid motions of a tetrahedron is isomorphic to a subgroup of S_4 .

Solution. Let G be the group of rigid motions of a tetrahedron with vertices labeled 1, 2, 3, 4. Then each $\alpha \in G$ sends some vertex to another vertex so that G acts on the set of vertices $A = \{1, 2, 3, 4\}$.

Since G acts on A , this gives rise to a homomorphism

$$\varphi : G \rightarrow S_4 \quad \text{where} \quad \varphi(\alpha)(i) = \alpha(i) \quad \text{for all} \quad i \in A.$$

To see that φ is injective, suppose $\varphi(\alpha)$ is the identity permutation in S_4 for some $\alpha \in G$. Then for every vertex $i \in A$, we have $\varphi(\alpha)(i) = i$, so that $\alpha(i) = i$. Since a rigid motion that fixes all vertices must be the identity rigid motion, then α is the identity element of G . Therefore, φ is injective and G is isomorphic to a subgroup of S_4 . ■

Exercise 1.7.21

Show that the group of rigid motions of a cube is isomorphic to S_4 . (This group acts on the set of four pairs of opposite vertices.)

Solution. Recall that $|G| = 24$ from [Exercise 1.2.10](#). Let G be the group of rigid motions of a cube, and let $A = \{a_1, a_2, a_3, a_4\}$ be the set of pairs of opposite vertices of the cube. Then each $\alpha \in G$ sends some pair of opposite vertices to another pair of opposite vertices so that G acts on A .

Using similar reasoning as in the exercise, we have an injective homomorphism

$$\varphi : G \rightarrow S_4 \quad \text{where} \quad \varphi(\alpha)(a_i) = \alpha(a_i) \quad \text{for all} \quad a_i \in A.$$

Since $|G| = 24 = |S_4|$, it follows that φ is surjective as well, hence $G \cong S_4$. ■

Exercise 1.7.22

Show that the group of rigid motions of an octahedron is isomorphic to a subgroup of S_4 . (This group acts on the set of four pairs of opposite faces.) Deduce that the groups of rigid motions of a cube and of an octahedron are isomorphic. (These two groups are isomorphic because these solids are “dual.”)

Solution. Let G be the group of rigid motions of an octahedron. Let $A = \{f_1, f_2, f_3, f_4\}$ be the set of pairs of opposite faces of the octahedron. Then each $\alpha \in G$ sends some pair of opposite faces to another pair of opposite faces so that G acts on A .

Using similar reasoning as in the previous exercises, we have an injective homomorphism

$$\varphi : G \rightarrow S_4 \quad \text{where} \quad \varphi(\alpha)(f_i) = \alpha(f_i) \quad \text{for all} \quad f_i \in A.$$

Therefore, G is isomorphic to a subgroup of S_4 . From [Exercise 1.2.11](#), we know that $|G| = 24$, hence $G \cong S_4$. From the previous exercise, we know that the group of rigid motions of a cube is also isomorphic to S_4 , so the groups of rigid motions of a cube and of an octahedron are isomorphic. ■

Exercise 1.7.23

Explain why the action of the group of rigid motions of a cube on the set of three pairs of opposite faces is not faithful. Find the kernel of this action.

Solution. The group of rigid motions of a cube has 24 elements, while permutations of the set of three pairs of opposing faces has 6 elements, so the action cannot be faithful as no homomorphism can be injective between finite groups of different sizes (if there was some injective homomorphism, it would imply bijectivity between the two groups of different cardinality, which is impossible).

Consider a cube such that its center is at the origin of $\mathbb{R}^3$ space. Then any rotation around the axes fixes a pair of faces, while sends the other two pairs back to themselves (to visualize this explicitly, consider the cube with vertices $(\pm 1, \pm 1, \pm 1)$ and the z -axis. A rotation around the z -axis would fix the faces $(\pm 1, \pm 1, 1)$ and $(\pm 1, \pm 1, -1)$, while it rotates the pair of faces $(1, \pm 1, \pm 1)$ and $(-1, \pm 1, \pm 1)$ back to each other—these are the faces that vary along the x -axis. One can construct the same for the pair of faces that lie along the y -axis and deduce the same thing). There are exactly 3 of these 180° rotations, so the kernel of this action consists of these rotations and the identity. ■